

B.Tech. I year, II sem
Physics –PH-113

Module-2: Electromagnetic Theory

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Delivered by:

Dr. Dheeraj Bhardwaj

Electromagnetism

Electromagnetism is a branch of physics that involves the study of electromagnetic forces, which is the combination of electric and magnetic forces. It is a type of interaction that occurs between electrically charged particles and responsible for all magnetic, electrical, and electromagnetic phenomena.

Before the invention of electromagnetism, people or scientists used to think electricity and magnetism are two different topics. But electricity and magnetism are inter related.

Static charge produces the electric field and moving charge produces magnetic field. Also time varying magnetic field produces electric field and time varying electric field is the source of magnetic field.

These observation laid a foundation for Electromagnetism.

Scalar and Vector Quantity

A **scalar** is a quantity that is fully described by a **magnitude only**.

A **vector** is a quantity that has both a **magnitude and a direction**.

Scalar Quantities

length, area, volume,
Speed, mass, density,
pressure, temperature,
energy



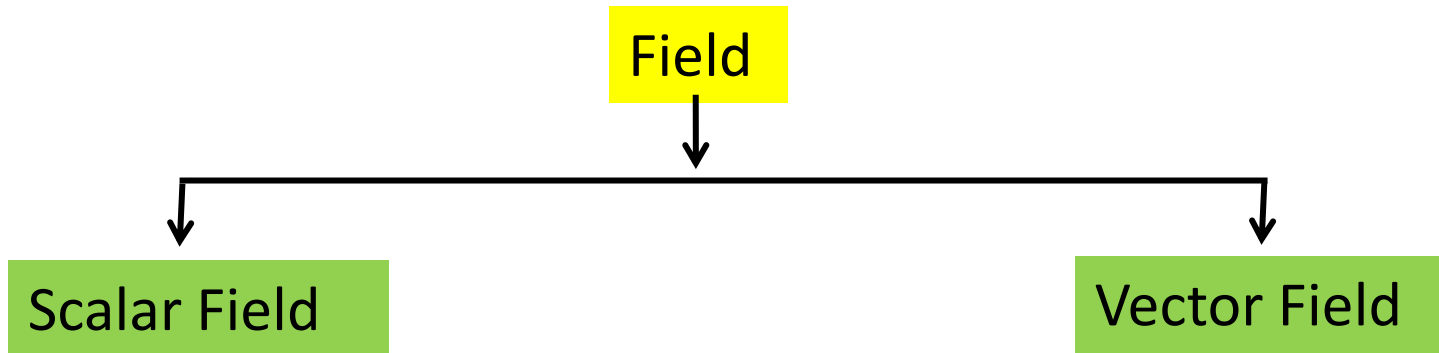
Vector Quantities

displacement,
velocity,
Acceleration,
Momentum, force



Scalar and Vector Fields

Field: The field or space in which the physical quantity depends upon the coordinates of the point of observation is called field of that physical quantity.



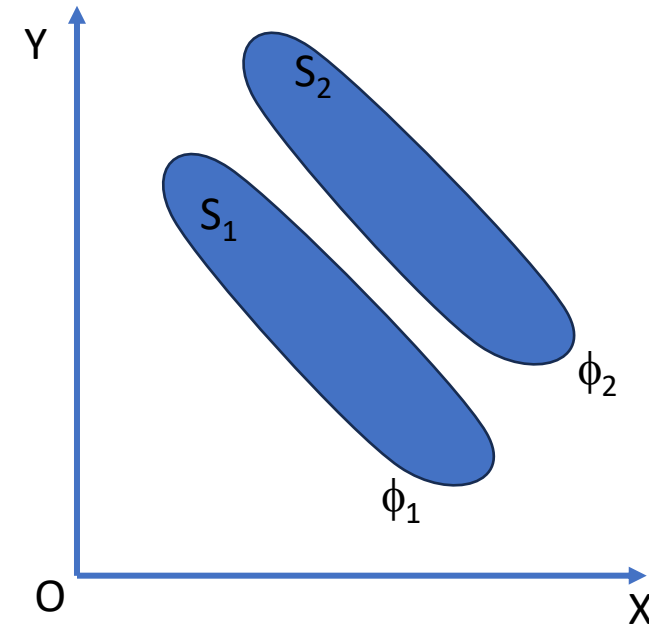
Scalar Field

- **Scalar field:** In space, when a scalar quantity is a function of coordinate of a point of observation is called scalar field.

A scalar field of a scalar quantity ϕ is defined at point $P(x,y,z)$

$$\phi = \phi(x, y, z)$$

- Scalar quantities are graphically represented by drawing surfaces at which the value of the scalar quantity remains a constant.
- Since one point in space corresponds to only one value of the scalar quantity i.e. scalar fields are single valued functions, hence these surfaces, called equiscalar surfaces, do not intersect with each other.



Equiscalar surfaces representing scalar quantity ϕ

Examples of Scalar Field:

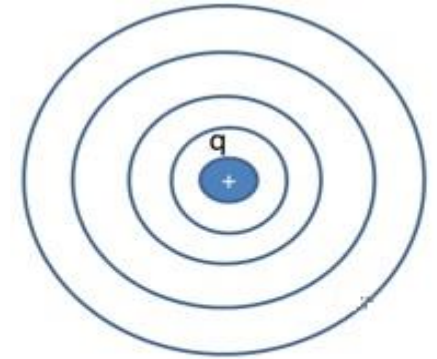
1. If we place a **point charge** at any place, then electric potential around it will depend on the position of the point.

$$V(r) = \frac{kq}{r} \quad \text{where} \quad r = \sqrt{x^2 + y^2 + z^2} \quad k = \frac{1}{4\pi\epsilon_0}$$

ϵ_0 is the permittivity of free space = 8.8542×10^{-12} F/m

Electric potential $V(r)$ is a scalar quantity

so field around the charge is called scalar potential field.



Equipotential surface for a single charge

2. **Gravitational Potential**

$$V(r) = \frac{GM}{r}$$

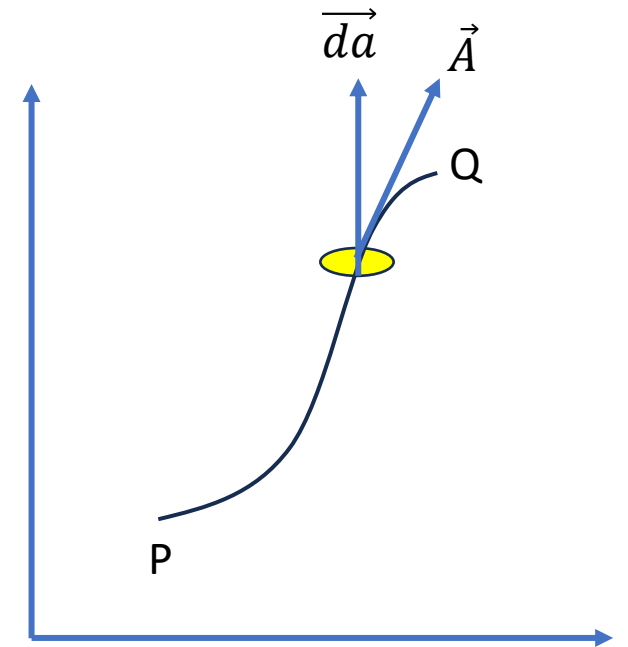
3. **Temperature $T(x, y, z)$**

Vector Field: In space, when a vector quantity is a function of coordinate of a point of observation is called a vector field.

The Vector field is expressed as

$$\vec{A} = A(x, y, z)$$

- Vector fields are graphically represented by lines. These lines are called field lines or flux lines.
- These lines are drawn in the field in such a way that the tangent at any point of the line gives the direction of the vector field at that point.
- The field lines drawn in electric and magnetic fields are also called lines of force.
- To express the magnitude of the vector field at any point of the field, an infinitesimal area element is drawn perpendicular to the field line. The number of field lines passing through the unit area, represents the magnitude of vector field.
- Hence, the magnitude of a vector field at any point of the field is equal to the number of field lines going out of the unit area taken normal to the direction of the vector at that point.



Representation of vector field

- Suppose the number of field lines or flux going out of an area element $\vec{da}$ perpendicular to the field lines at any point is $d\phi$. Then the magnitude of the intensity of the vector field will be.

$$|\vec{A}| = \frac{d\phi}{da} \quad (...1)$$

∴ Vector field

$$\vec{A} = \left(\frac{d\phi}{da}\right) \hat{n} \quad (...2)$$

Where $\hat{n}$ is the unit vector perpendicular to the area da

$$\hat{n} da = \vec{da} \quad (...3)$$

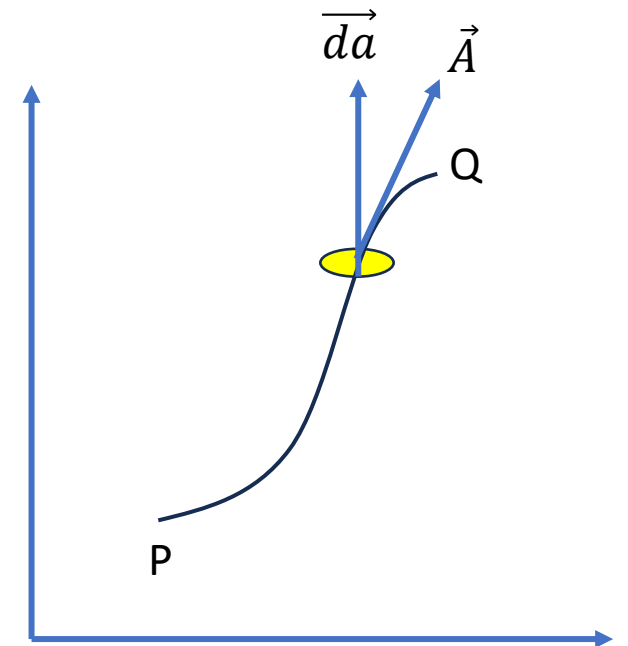
Taking the dot product of unit vector $\hat{n}$ with the vector field $\vec{A}$,

$$\vec{A} \cdot \hat{n} = \left(\frac{d\phi}{da}\right) \hat{n} \cdot \hat{n} = \frac{d\phi}{da}$$

$$\text{or } d\phi = (\vec{A} \cdot \hat{n}) da = \vec{A} \cdot \hat{n} da$$

$$\text{Flux } d\phi = \vec{A} \cdot \vec{da} \quad (...4)$$

Like scalar fields, positional function of a vector field is also a single valued function i.e. a vector field lines do not intersect with each other.



Representation of vector field

Examples:

1. Electric field

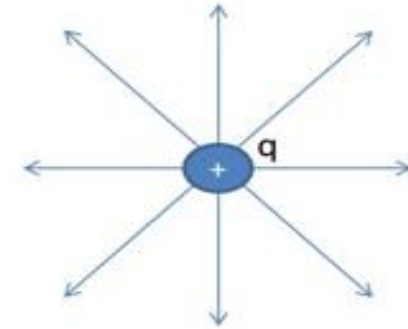
$$\vec{E} = \frac{kq}{r^2} \hat{r}$$

2. Magnetic field

$$d\vec{B} = \frac{\mu_o Idl \sin \theta}{4\pi r^2} \hat{r}$$

3. Gravitational field.

$$\vec{g} = -\frac{GM}{r^2} \hat{r}$$



Electric field lines for a single positive charge

DEL OPERATOR or GRADIENT OPERATOR (∇) :-

- The Del, or nabla, is an operator used in mathematics (particularly in vector calculus) as a vector differential operator, usually represented by the nabla symbol ∇ .
- In Cartesian coordinates Del is written by

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

- When del operator applied to a function defined on a one-dimensional domain, it denotes the standard derivative of the function as defined in calculus.
- This vector differential operator, otherwise known as the **gradient operator**, is not a vector itself, but when it operates on a scalar function.

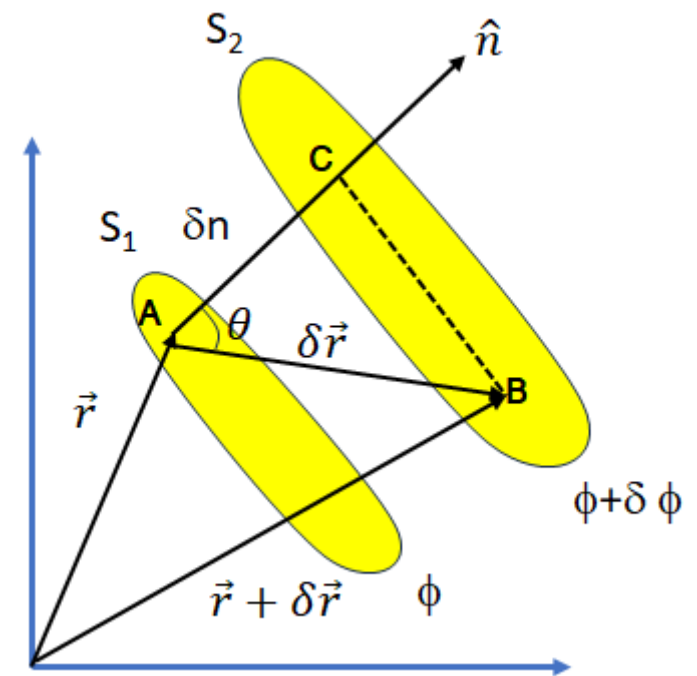
- **The operator is useful in defining**

1. The gradient of a scalar V , written as =>
2. The divergence of a vector A , written as =>
3. The curl of a vector A , written as =>
4. The Laplacian of a scalar V , written as =>

- | | |
|-------------------|------------------|
| ∇V | (grad V) |
| $\nabla \cdot A$ | (div A) |
| $\nabla \times A$ | (Curl A) |
| $\nabla^2 V$ | (Laplacian V) |

Gradient of Scalar Field

- Suppose in a scalar field two surfaces S_1 and S_2 represent surfaces with constant scalar quantities ϕ and $\phi + \delta\phi$ respectively. The position vector of any point A on S_1 relative to the origin O is $\vec{r}$ and the position vector of another point B on Surface two is $\vec{r} + \delta\vec{r}$.
- $\therefore$ displacement of B with respect to A is $AB = \delta\vec{r}$ and the rate of change of the scalar function in the direction of AB = $\frac{\delta\phi}{\delta r}$ (...1)
- Now we draw a perpendicular AC from A on surface S_2 . The minimum distance $AC = \delta n$ between the surface S_1 and S_2 will be in the direction of their normal.
- $\therefore$ The rate of change of scalar function ϕ in the direction of AC = $\frac{\delta\phi}{\delta n}$ (...2)



Gradient of a scalar quantity

Gradient of Scalar Field

If the distance δr is infinitesimal, then $\triangle ABC$ can be considered a right-angled triangle. Suppose the angle between $\delta \vec{r}$ and $\delta \vec{n}$ is θ , then

$$AC = AB \cos \theta$$

$$\text{or } \delta n = \delta r \cos \theta \quad (...3)$$

Substituting the value of δr in equation (1), we get

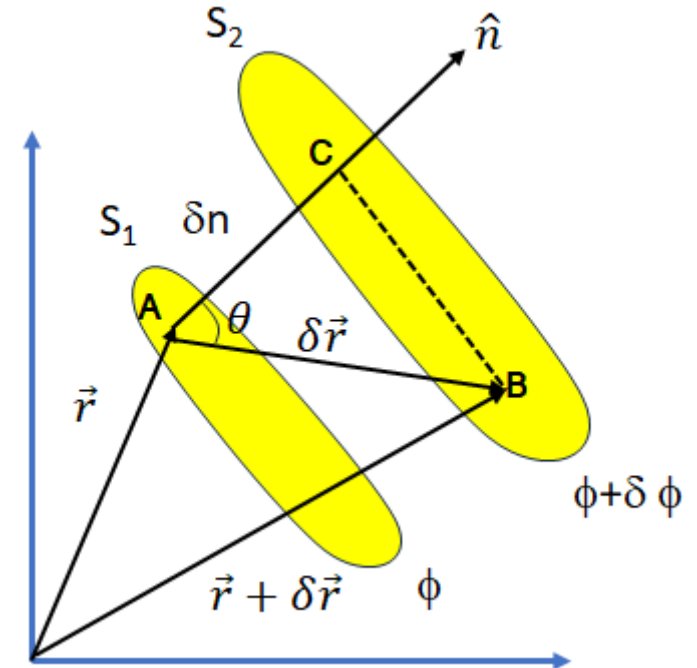
Rate of change of ϕ in the direction of AB

$$= \frac{\delta \phi}{\delta n} \cos \theta$$

As the maximum value of $\cos \theta = 1$

$\therefore$ The maximum rate of change of ϕ

$$= \left(\frac{\delta \phi}{\delta r} \right)_{\max} = \left(\frac{\delta \phi}{\delta n} \right)$$



Gradient of a scalar quantity

Gradient of Scalar Field

In the limit $\delta r \rightarrow 0$

$$\left(\frac{d\phi}{dr}\right)_{max} = \left(\frac{d\phi}{dn}\right)$$

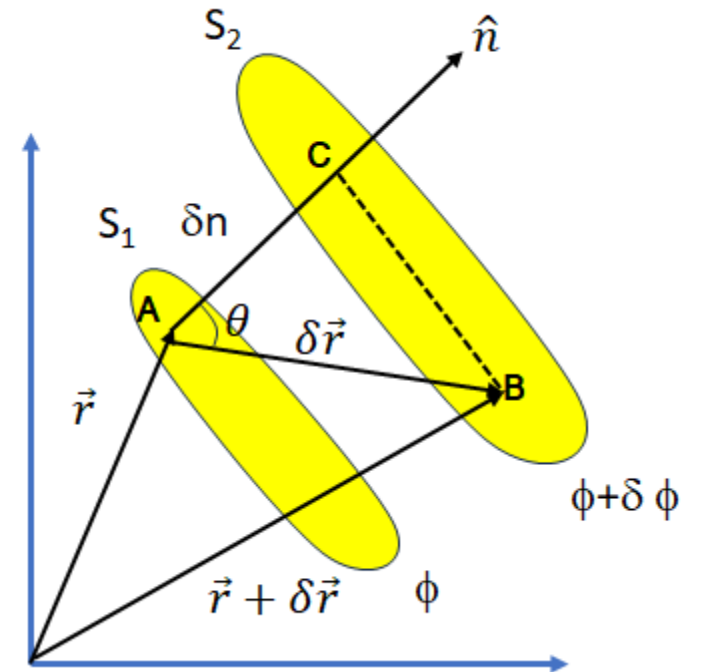
Hence, the rate of change of scalar function is maximum in a direction normal to the equi scalar surface.

In a scalar field, the maximum rate of change of scalar function in space is known as gradient of the scalar field.

It is a vector quantity and its direction is normal to the scalar surface, that is the direction in which the rate of change is maximum.

If we take $\hat{n}$ as unit vector normal to the equiscalar surface, then

$$\text{Gradient } \phi = \left(\frac{d\phi}{dn}\right) \hat{n} = \left(\frac{d\phi}{dr}\right)_{max} \hat{n}$$



Gradient of Scalar Field

- The gradient of a scalar field at any point is the maximum rate of change of the field at that point.
The **gradient** of a scalar field V is a vector that represents both the magnitude and the direction of the maximum space rate of increase of V .
- It is a vector quantity, and its direction is normal to the equiscalar surface i.e. the direction in which rate of change is maximum.

Example: Electric field is related to the electric potential V by

$$\vec{E} = -\nabla V$$

If $\phi(x, y, z)$ is a scalar field, then gradient of ϕ is defined as

$$\text{grad } V = \nabla V = \left(\hat{i} \frac{\partial V}{\partial x} + \hat{j} \frac{\partial V}{\partial y} + \hat{k} \frac{\partial V}{\partial z} \right)$$

where
$$\nabla = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right)$$

Gradient of Scalar Field contd..

A mathematical expression for the gradient can be obtained by evaluating the difference in the field dV between points P_1 and P_2 of Figure, where V_1 , V_2 , and V_3 are contours on which V is constant. From calculus,

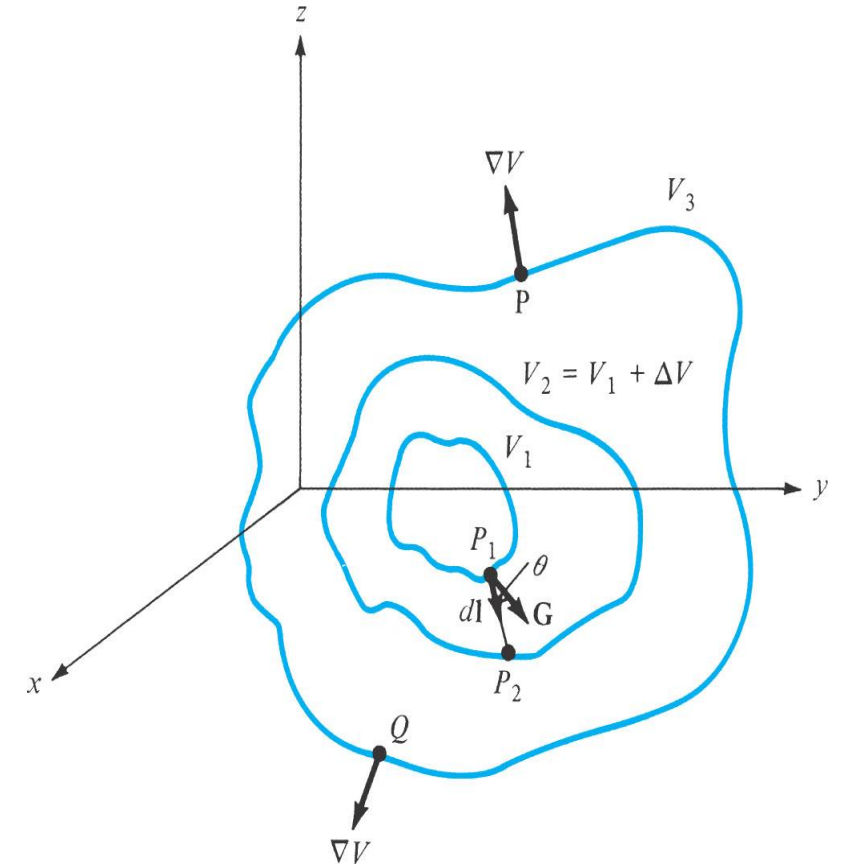
$$\begin{aligned} dV &= \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz \\ &= \left(\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \right) \cdot (dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z) \end{aligned} \quad \dots(1)$$

For convenience, let

$$\bullet \quad \mathbf{G} = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \quad \dots(2)$$

Then

$$dV = \mathbf{G} \cdot d\mathbf{l} = G \cos \theta \, dl$$



Gradient of Scalar Field contd..

Or

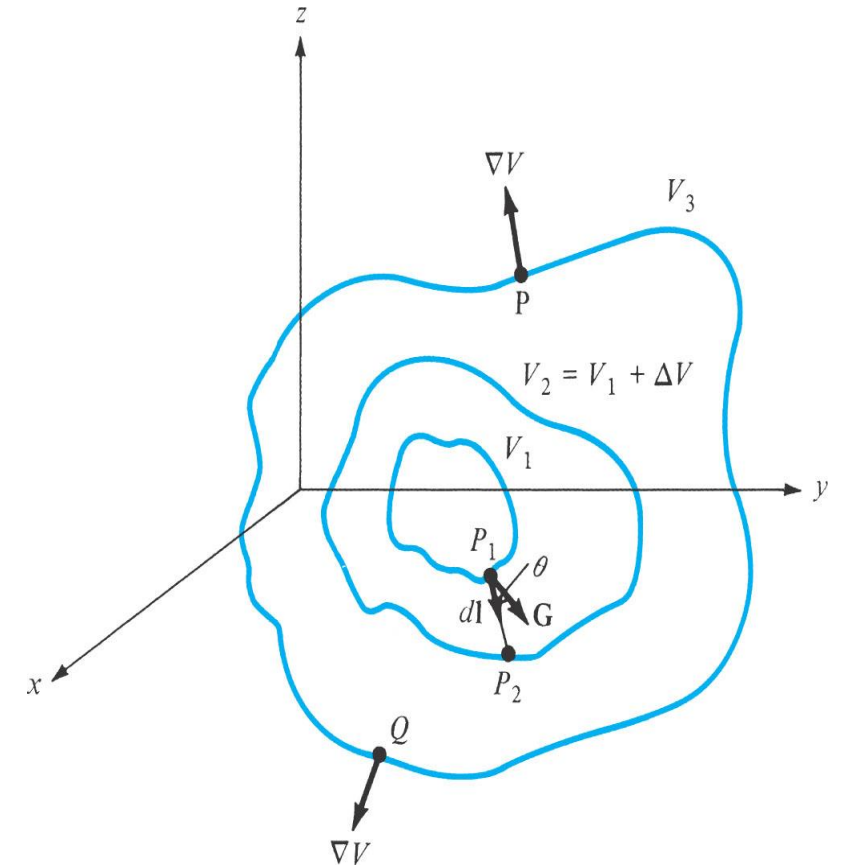
$$\frac{dV}{dl} = G \cos \theta \quad \dots(3)$$

where $d\mathbf{l}$ is the differential displacement from P_1 to P_2 and θ is the angle between $\mathbf{G}$ and $d\mathbf{l}$. From eq. (3), we notice that dV/dl is a maximum when $\theta=0$, that is, when $d\mathbf{l}$ is in the direction of $\mathbf{G}$. Hence,

$$\left. \frac{dV}{dl} \right|_{\max} = \frac{dV}{dn} = G \quad \dots(4)$$

where dV/dn is the normal derivative. Thus G has its magnitude and direction as those of the maximum rate of change of V . By definition, $\mathbf{G}$ is the gradient of V . Therefore:

$$\text{grad } V = \nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \quad \dots(5)$$



Gradient of Scalar Field contd..

The gradient of V can be expressed in Cartesian coordinates as

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

for cylindrical coordinates,

- $$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z \quad \dots(6)$$

- and for spherical coordinates,

- $$\nabla V = \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \quad \dots(7)$$

The following computation formulas on gradient, which are easily proved, should be noted:

$$(i) \quad \nabla(V + U) = \nabla V + \nabla U \quad \dots(8 a)$$

$$(ii) \quad \nabla(VU) = V\nabla U + U\nabla V \quad \dots(8 b)$$

$$(iii) \quad \nabla\left[\frac{V}{U}\right] = \frac{U\nabla V - V\nabla U}{U^2} \quad \dots(8 c)$$

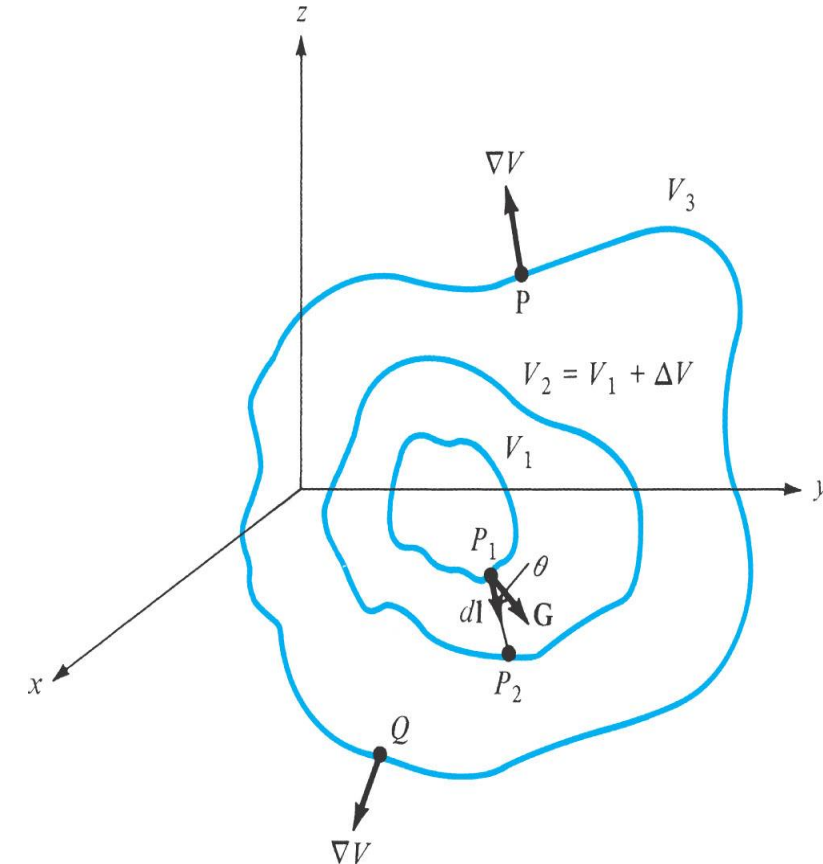
$$(iv) \quad \nabla V^n = nV^{n-1} \nabla V \quad \dots(8 d)$$

where U and V are scalars and n is an integer

Gradient of Scalar Field contd..

Also take note of the following fundamental properties of the gradient of a scalar field V :

- The magnitude of ∇V equals the maximum rate of change in V per unit distance.
- ∇V points in the direction of the maximum rate of change in V .
- ∇V at any point is perpendicular to the constant V surface that passes through that point (see points P and Q in figure).
- The projection (or component) of ∇V in the direction of a unit vector $\mathbf{a}$ is $\nabla V \cdot \mathbf{a}$ and is called the directional derivative of V along $\mathbf{a}$. This is the rate of change of V in the direction of $\mathbf{a}$. For example, dV/dl in eq. (3) is the directional derivative of V along P_1P_2 in Figure. Thus the gradient of a scalar function V provides us with both the direction in which V changes most rapidly and the magnitude of the maximum directional derivative of V .
- If $\mathbf{A} = \nabla V$, V is said to be the scalar potential of $\mathbf{A}$



Gradient of Scalar Field contd..

Example:

Find the gradient of the following scalar fields:

(a) $V = e^{-z} \sin 2x \cosh y$

(b) $U = \rho^2 z \cos 2\phi$

(c) $W = 10r \sin^2 \theta \cos \phi$

Solution:

$$\begin{aligned} \text{(a) } \nabla V &= \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \\ &= 2e^{-z} \cos 2x \cosh y \mathbf{a}_x + e^{-z} \sin 2x \sinh y \mathbf{a}_y - e^{-z} \sin 2x \cosh y \mathbf{a}_z \end{aligned}$$

$$\begin{aligned} \text{(b) } \nabla U &= \frac{\partial U}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial U}{\partial \phi} \mathbf{a}_\phi + \frac{\partial U}{\partial z} \mathbf{a}_z \\ &= 2\rho z \cos 2\phi \mathbf{a}_\rho - 2\rho z \sin 2\phi \mathbf{a}_\phi + \rho^2 \cos 2\phi \mathbf{a}_z \end{aligned}$$

$$\begin{aligned} \text{(c) } \nabla W &= \frac{\partial W}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial W}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial W}{\partial \phi} \mathbf{a}_\phi \\ &= 10 \sin^2 \theta \cos \phi \mathbf{a}_r + 10 \sin 2\theta \cos \phi \mathbf{a}_\theta - 10 \sin \theta \sin \phi \mathbf{a}_\phi \end{aligned}$$

Flux of a Field

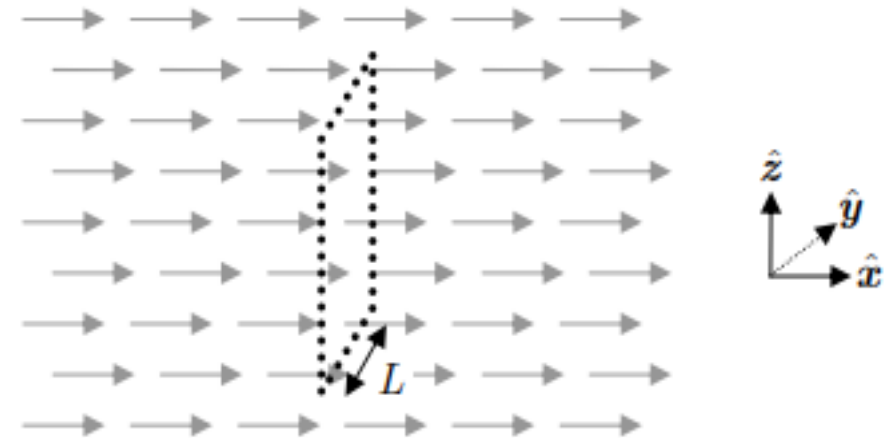
- The term flux has been used to study the flowing field.
- Flux is the amount of vector field passing through a surface per second.
- The **total flux** depends on strength of the field, the size of the surface it passes through, and their orientation.

- Consider a region of space in which there is a constant vector field,

$$\vec{E}(x, y, z) = a\hat{x}$$

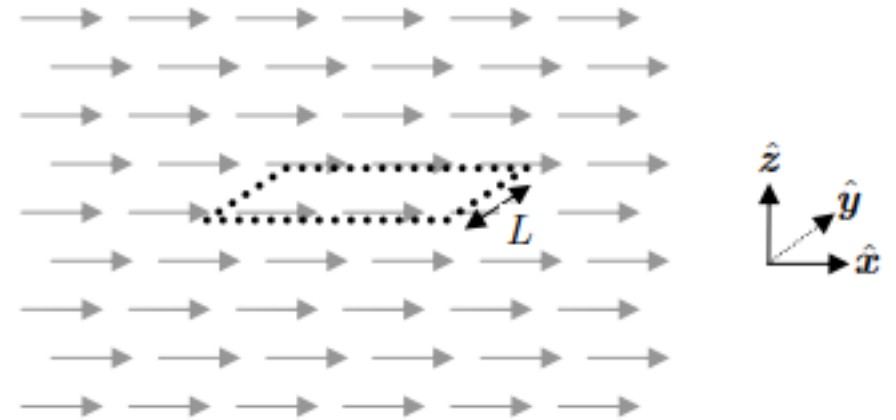
- Imagine that the vector field is a fluid flowing through space.
- Then, the flux of the field through an area is the amount of “fluid” flowing through that area. Now, in this case, the area through which fluid is flowing is L^2 , and the field strength is “a”, and so

$$\text{Flux} = aL^2$$



Flux of a Field....

- Now, consider a slightly different example.
- In this case, it should be pretty clear that none of the fluid is passing through the square, because the plane of the square is parallel to the fluid flow.



$$\boxed{\text{Flux} = 0}$$

Flux of a Field....

- If the area of the square is inclined at some angle to the fluid flow.
- The field vectors can be broken down into two components.
- Now, let's have a look at the flux of each of these components through the square:
- (i) The $a \sin \theta$ component is parallel to the plane, and therefore, as in the second example above, it will produce 0 flux.
- (ii) The $a \cos \theta$ component is perpendicular to the plane, and therefore, as in the first example above, it will produce a flux $aL^2 \cos \theta$
- Adding these two fluxes together

$$\text{Flux} = aL^2 \cos \theta$$

The general expression for flux is

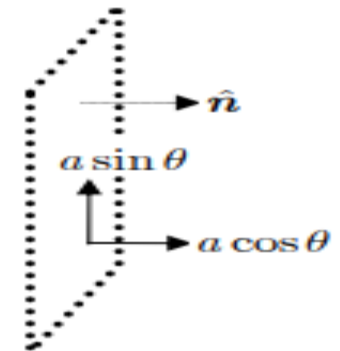
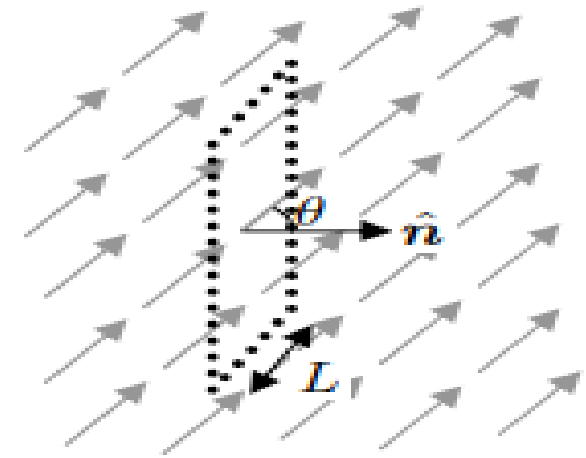
$$\text{Flux } \phi = (\mathbf{A} \cdot \mathbf{n}') \times \text{Area}$$

$$\phi = (\mathbf{A} \cdot \mathbf{n}) \, ds$$

Where $\hat{n}$ is a unit vector that points in a direction perpendicular to the area ds .

$$\text{Flux} = (\mathbf{E} \cdot \hat{n}) \times \text{Area}$$

$$\Phi = \mathbf{E} \cdot \hat{n} A$$



Divergence of Vector Field

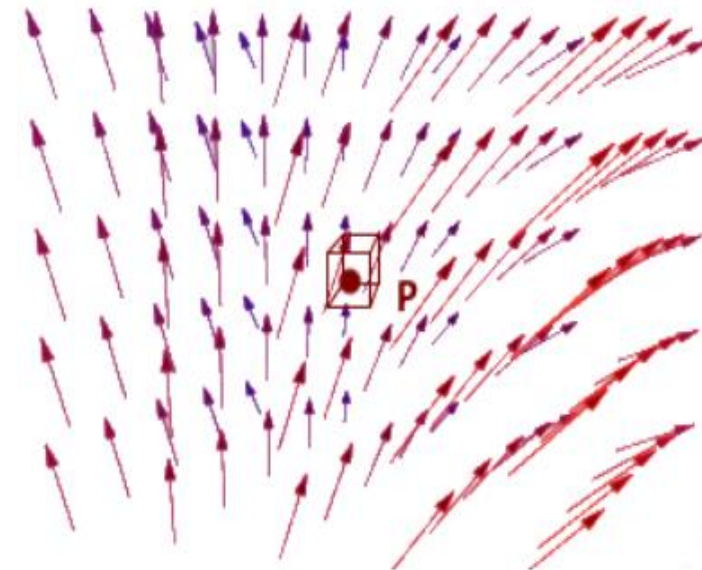
- The net outflow of the flux of a vector field $\mathbf{A}$ from a closed surface S is obtained from the integral $\oint_S \vec{A} \cdot \vec{dS}$. We now define the divergence of $\mathbf{A}$ as the net outward flow of flux per unit volume over a closed incremental surface.
- The **divergence** of $\mathbf{A}$ at a given point P is the *outward* flux per unit volume as the volume shrinks about P .

Hence,

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v} \quad (\dots 1)$$

where Δv is the volume enclosed by the closed surface S in which P is located.

- Physically, we may regard the divergence of the vector field $\mathbf{A}$ at a given point as a measure of how much the field diverges or emanates or originates from that point.



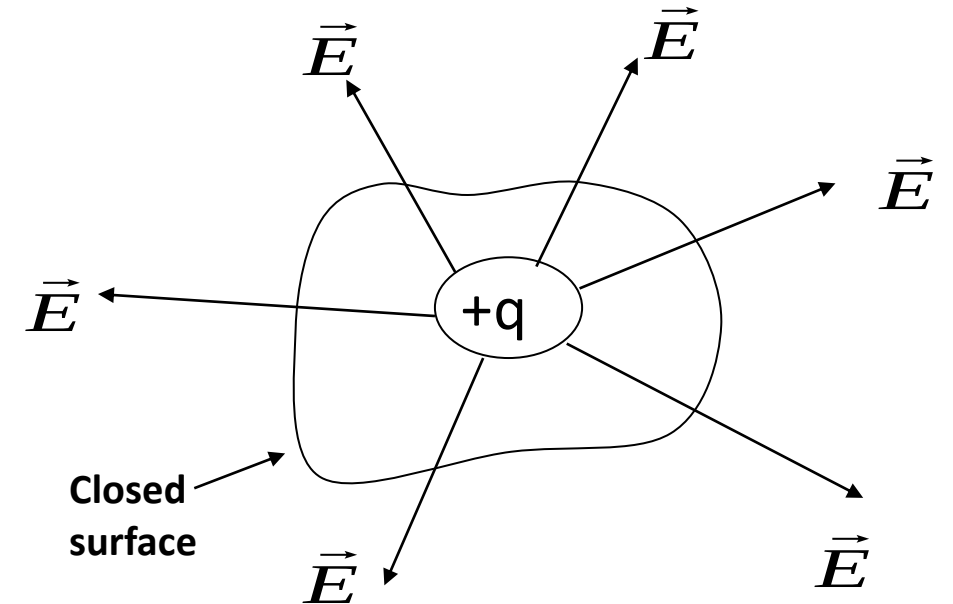
Divergence of Vector Field.....

Physical Significance:

1. If +q charge is placed inside a volume.
 - The net flux is emerging out of the closed surface.

$$\text{Div}\vec{E} = +ve$$

- Source of electric field lines placed inside a given volume.



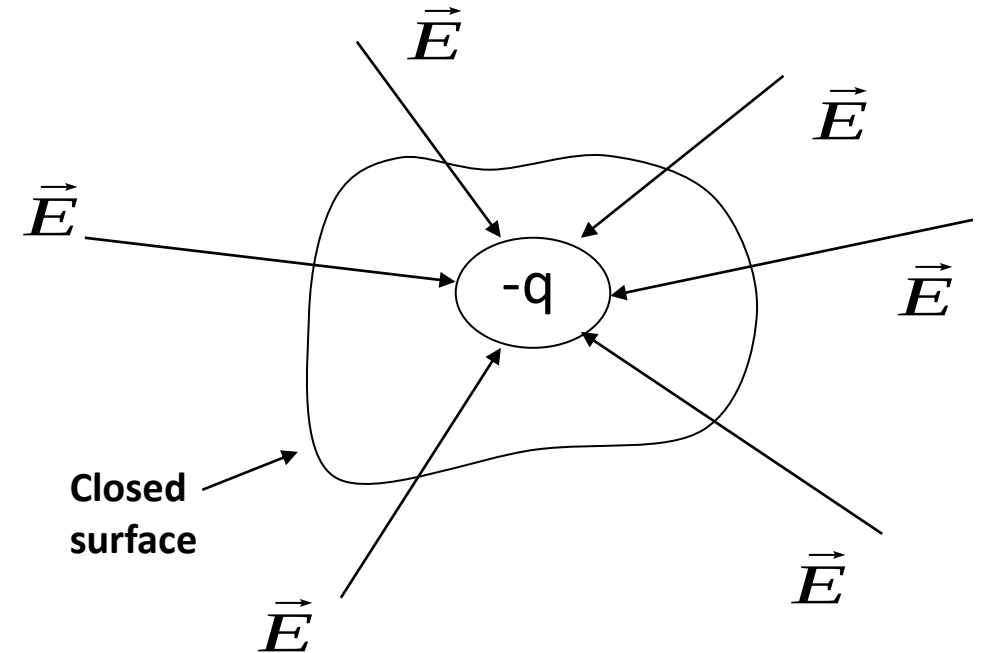
Divergence of Vector Field.....

2. If $-q$ charge placed inside a volume.

- The Electric field lines converge toward $-q$ charge the closed volume.

$$\text{Div}\vec{E} = -ve$$

- Sink of electric field lines placed inside a given volume.



Divergence of Vector Field.....

3. If +q charge is placed outside from closed surface net flux passing is zero i.e. net incoming flux will be equal to net emerging flux.

$$\text{Div} \vec{E} = 0$$

- No source or sink is present inside the volume.

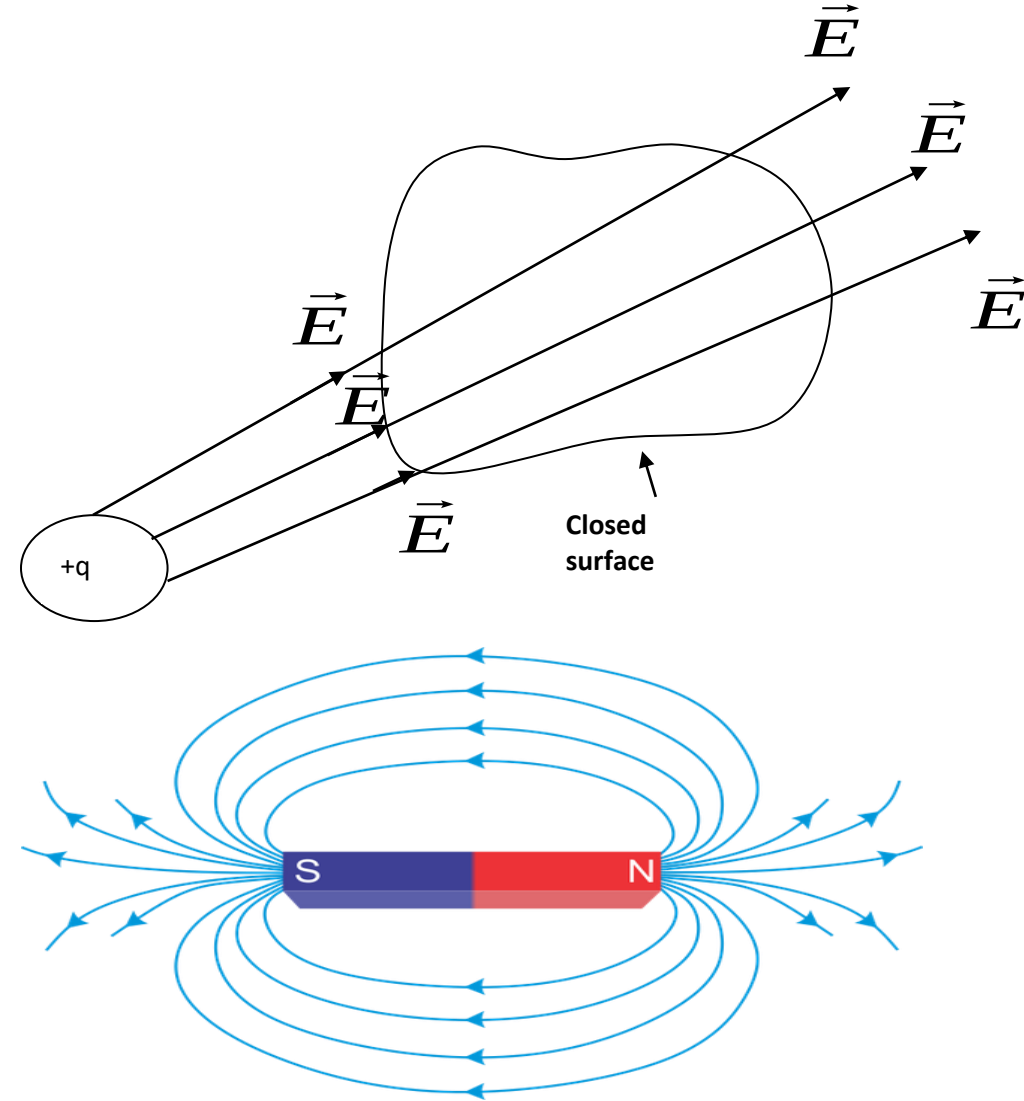
Example: Magnetic lines of force always form close loop i.e. no. of magnetic field lines entering into a given volume is equal to no. of field lines emerging.

So

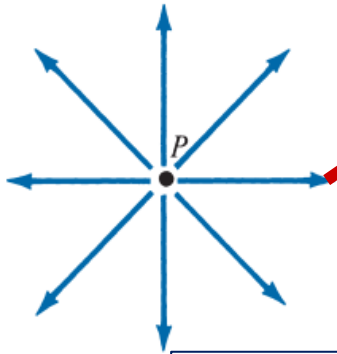
$$\text{Div} \vec{B} = 0$$

Vector field whose divergence is zero is called Solenoidal vector field.

i.e. if $\text{Div} \vec{A} = 0$, the $\vec{A}$ is a Solenoidal vector.

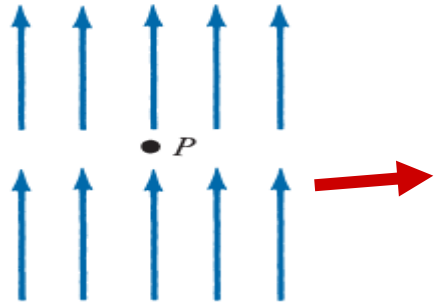
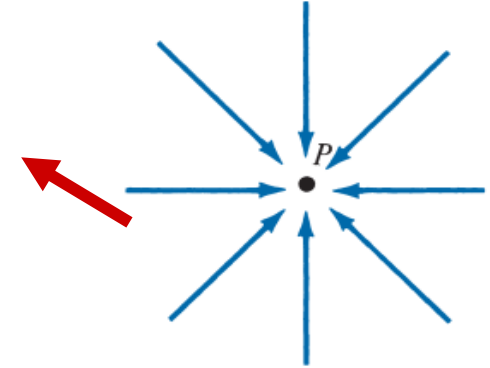


Divergence of Vector Field



This figure shows that the divergence of a vector field at point P is positive because the vector diverges (or spreads out) at P .

This figure shows that the divergence of a vector field at point P is negative because the vector converges at P .

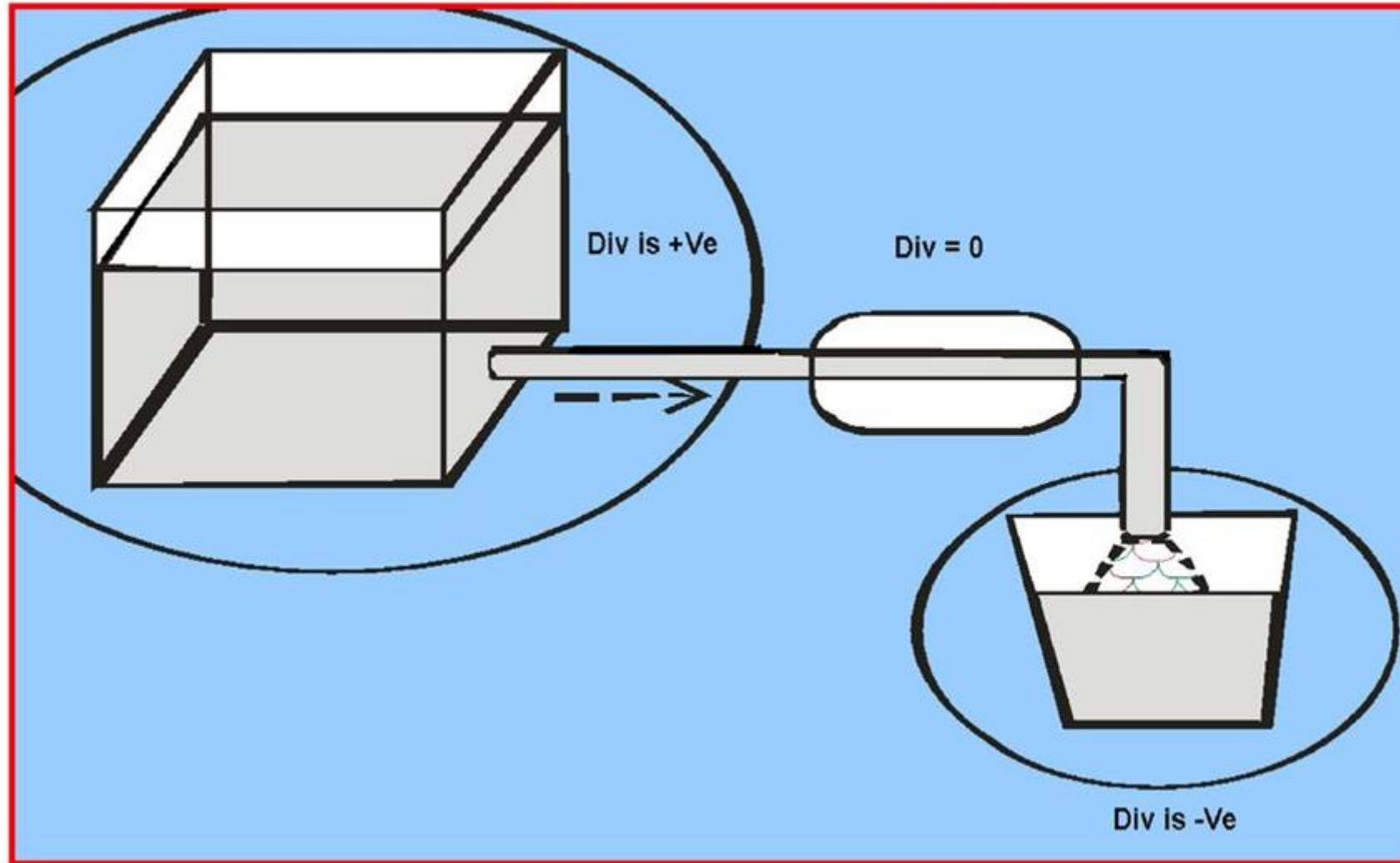


This figure shows that the divergence of a vector field at point P is zero at P .

The divergence of a vector field can also be viewed as simply the limit of the field's source strength per unit volume (or source density); it is positive at a source point in the field, and negative at a sink point, or zero where there is neither sink nor source.

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

Example for Divergence



Divergence of Vector Field in Cartesian Coordinates

According to the definition of divergence

$$\text{div } \vec{A} = \lim_{dV \rightarrow 0} \left[\frac{1}{dV} \oint_S \vec{A} \cdot d\vec{a} \right] \quad \dots(5)$$

Comparing equation (4) and (5),

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Therefore, the divergence of a vector function is equal to the scalar product of the del operator and the vector function.

Gauss Divergence Theorem

- To prove the divergence theorem, subdivide volume V enclosed by a closed surface S in a vector field $\vec{A}$ into a large number of infinitesimally small rectangular parallelepipeds. If the k th cell has volume dV_k and is bounded by surface S_k

$$\oint_S \vec{A} \cdot d\vec{a} = \sum_k \oint_{S_k} \vec{A}_k \cdot d\vec{a}_k \quad \dots(6)$$

But the total flux emerging from the faces of the infinitesimally small volume elements is (from equation (4))

$$\oint_{S_k} \vec{A}_k \cdot d\vec{a}_k = (\nabla \cdot \vec{A}_k) dV_k \quad \dots(7)$$

Substituting equation (7) in (6)

$$\oint_S \vec{A} \cdot d\vec{a} = \sum_k (\nabla \cdot \vec{A}_k) dV_k$$

Gauss Divergence Theorem

- If $dV_k \rightarrow 0$ then $\oint_S \vec{A} \cdot d\vec{a} = \int_v (\nabla \cdot \vec{A}) dV$ (8)

This theorem transforms the volume integral into the surface integral.

The **divergence theorem** states that the total outward flux of a vector field **A** through the *closed* surface *S* is the same as the volume integral of the divergence of **A**.

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_v \nabla \cdot \mathbf{A} dv$$

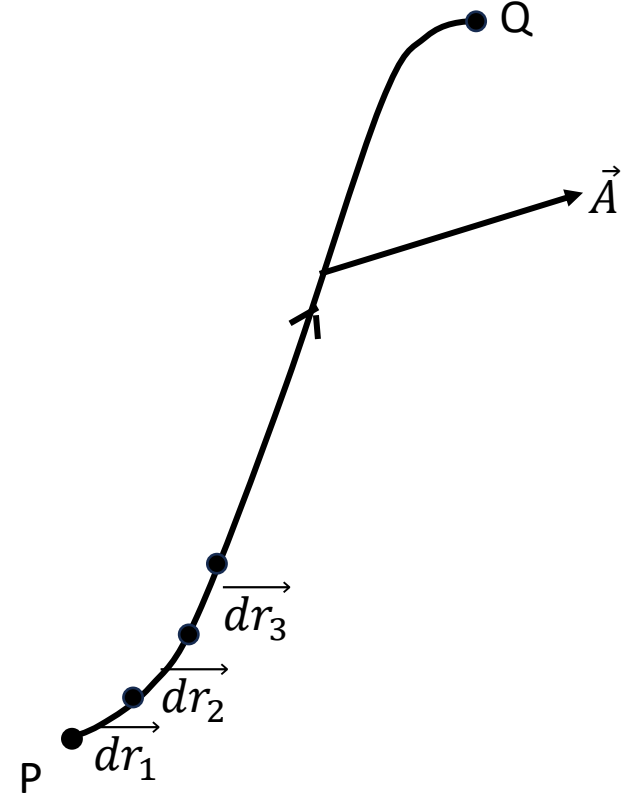
Physical Importance of Line Integral

- If in a vector field, the line integral of the field between two points does not depend on the path but depends only on the positions of the end points, then such a field is known as a conservative field.
- Electric field and Gravitational field are examples of conservative fields.
- If a vector $\vec{A}$ represents a force acting during a given displacement $\vec{dr}$, then $\vec{A} \cdot \vec{dr}$ will represent the work done by the force for the displacement $\vec{dr}$. Hence the total work done by the force field for the displacement from the point P to point Q is $\int_P^Q \vec{A} \cdot \vec{dr}$.
- If vector $\vec{A}$ represents electric field $\vec{E}$ Then $\int_P^Q \vec{E} \cdot \vec{dr}$ will represent the work done in taking unit positive charge from the point P to Q because electric field intensity is equal to the force acting on unit positive charge. This work done against electric field is called potential difference.

$$\Delta V = V_Q - V_P = - \int_P^Q \vec{E} \cdot \vec{dr}$$

If in any circuit a source (cell or battery) is used then line integral on complete path $\oint \vec{E} \cdot \vec{dl}$ is equal to the electromotive force of the source

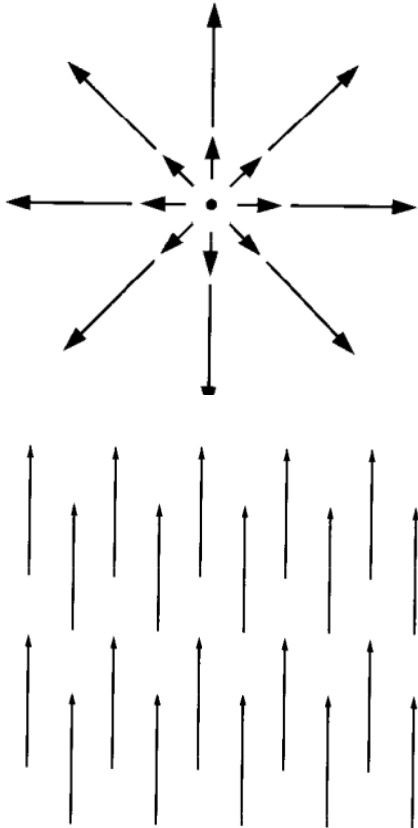
i.e. Electromotive Force (E.M.F) = $\oint \vec{E} \cdot \vec{dl}$



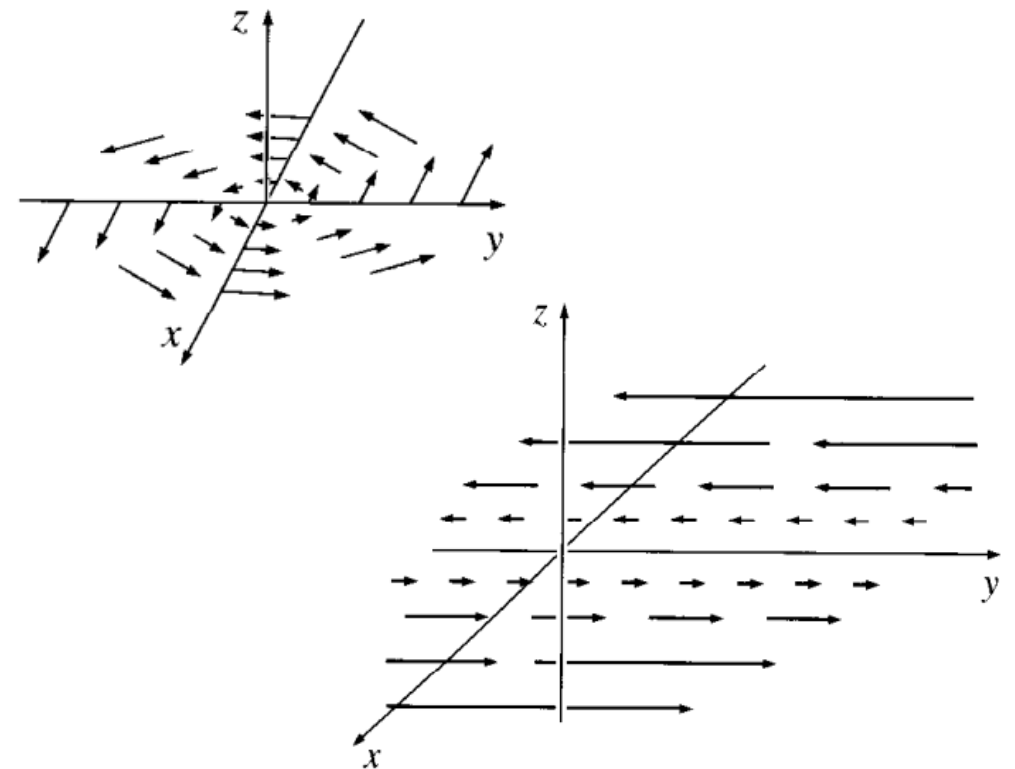
Curl of a Vector Field

The curl of a vector field is a measure of how much the vector curls around the point of observation.

Curl of vector field is zero



Curl of vector field is not zero



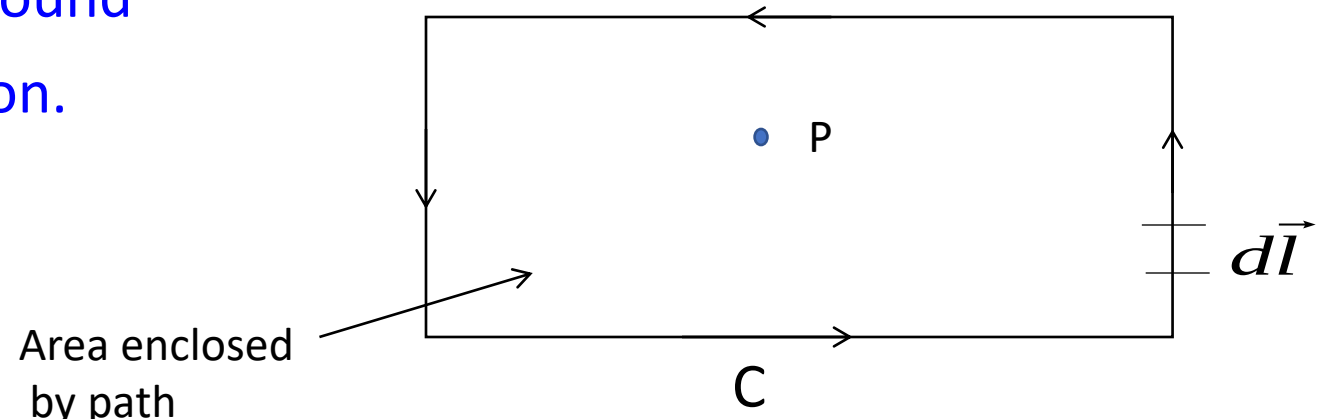
Curl of a Vector Field

- Curl of any vector field is related to line integral of that vector.
- Curl of a vector field is a vector quantity.

Consider a point P in a vector field $\vec{A}$ and path C enclose that point. $d\vec{l}$ is a small length element on this path. If area enclosed by the closed path is small so that vector field $\vec{A}$ same at each point of path then

$\oint_C \vec{A} \cdot d\vec{l}$ = line integral of vector field over a closed path.

- The line integral of a vector function around a closed path is also known as circulation.

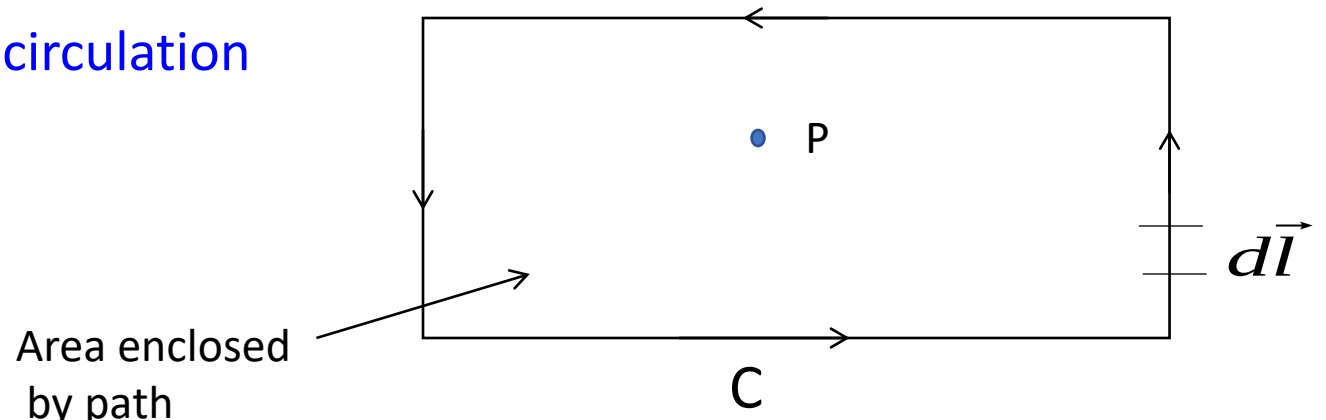


Curl of a Vector Field

- The line integral depends on the orientation of the surface enclosed by the closed path.
- If the surface enclosed by the closed path is kept in such a way that the line integral or circulation becomes maximum, then in this orientation the ratio of the line integral to the area enclosed by the path with the area tending to zero, is equal to the magnitude of the curl of the vector field at the position of the area element.

$$\text{curl} \vec{A} \cdot \hat{n} = \lim_{da \rightarrow 0} \left[\frac{\oint_C \vec{A} \cdot d\vec{l}}{da} \right]$$

- Its direction is perpendicular to the plane which contains the area element with maximum circulation around its boundary.



Curl of a Vector Field....

or

$$\text{curl} \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Physical significance of curl:

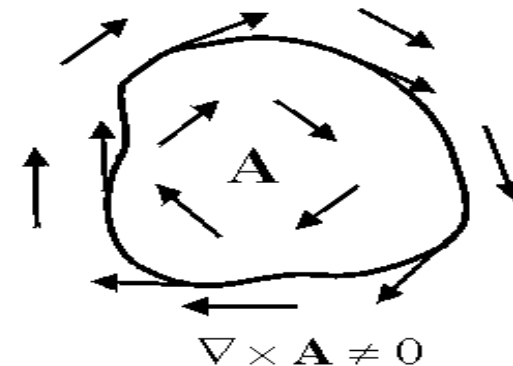
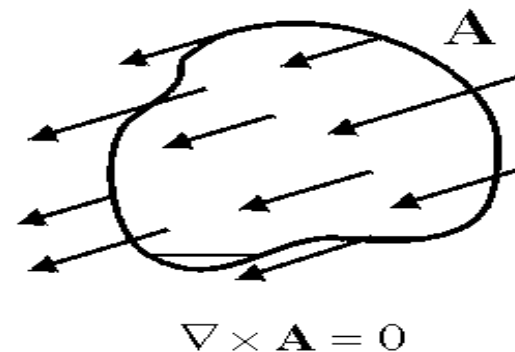
Curl of a vector field measure how much the vector $\vec{A}$ “curl around “ the point of observation. It comes from its rotational motion.

Example: For static field

$$\nabla \times \vec{E} = 0 \quad (\text{Maxwell III equation})$$

$$\nabla \times \vec{B} = \mu_o J \quad (\text{Maxwell IV equation})$$

If curl of a vector field is zero, these vectors are called irrotational vector field.



Curl of a Vector field

The **curl** of a vector field **A** is an axial (or rotational) vector whose magnitude is the maximum circulation of **A** per unit area as the area tends to zero and whose direction is the normal direction of the area when the area is oriented to make the circulation maximum.

That is

$$\text{curl } \mathbf{A} = \nabla \times \mathbf{A} = \left(\lim_{\Delta S \rightarrow 0} \frac{\oint_L \mathbf{A} \cdot d\mathbf{l}}{\Delta S} \right)_{\max} \mathbf{a}_n \quad (...1)$$

where the area ΔS is bounded by the curve L and $\mathbf{a}_n$ is the unit vector normal to the surface ΔS and is determined by using the right-hand rule.

In Cartesian coordinates the curl of **A** is easily found using

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

or

$$\nabla \times \mathbf{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \mathbf{a}_x + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] \mathbf{a}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] \mathbf{a}_z \quad (...2)$$

Curl of a Vector field

In Cylindrical coordinates the curl of $\mathbf{A}$ is

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

or

$$\nabla \times \mathbf{A} = \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \mathbf{a}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right] \mathbf{a}_\phi + \frac{1}{\rho} \left[\frac{\partial(\rho A_\phi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right] \mathbf{a}_z \quad (...3)$$

In Spherical coordinates the curl of $\mathbf{A}$ is

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{a}_r & r \mathbf{a}_\theta & r \sin \theta \mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

or

$$\nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left[\frac{\partial(A_\phi \sin \theta)}{\partial \theta} - \frac{\partial A_\theta}{\partial \phi} \right] \mathbf{a}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial(r A_\phi)}{\partial r} \right] \mathbf{a}_\theta + \frac{1}{r} \left[\frac{\partial(r A_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right] \mathbf{a}_\phi$$

(...4)

Curl of a Vector field

Note the following properties of the curl:

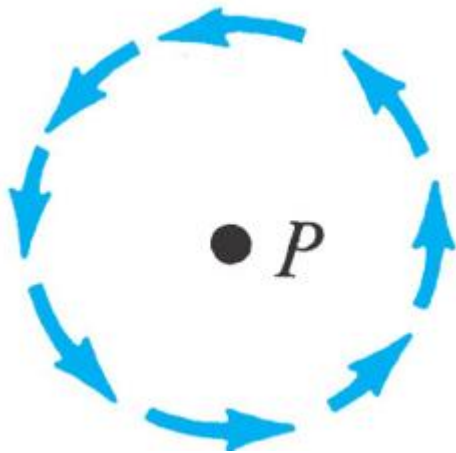
1. The curl of a vector field is another vector field.
2. $\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}$
3. $\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$
4. $\nabla \times (V\mathbf{A}) = V\nabla \times \mathbf{A} + \nabla V \times \mathbf{A}$
5. The divergence of the curl of a vector field vanishes; that is, $\nabla \cdot (\nabla \times \mathbf{A}) = 0$.
6. The curl of the gradient of a scalar field vanishes; that is, $\nabla \times \nabla V = \mathbf{0}$ or $\nabla \times \nabla = \mathbf{0}$.

Curl of a Vector field

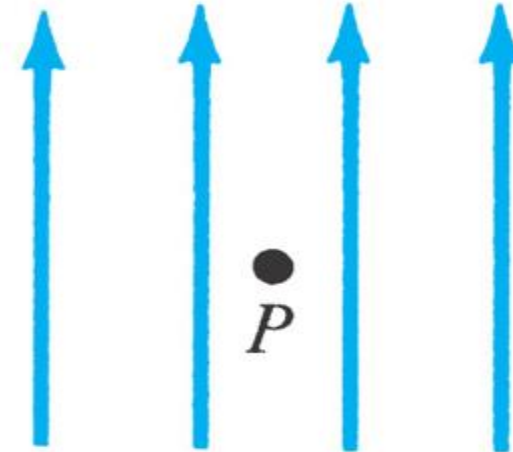
Physical Significance:

The curl provides the maximum value of the circulation of the field per unit area (or circulation density) and indicates the direction along which this maximum value occurs. The curl of a vector field $\mathbf{A}$ at a point P may be regarded as a measure of the circulation or how much the field curls around P . If the curl of a vector field is zero, these vectors are called an irrotational vector field.

For example.



This figure shows that the curl of a vector field around P is non-zero and directed out of the page.



This figure shows a vector field with zero curl.

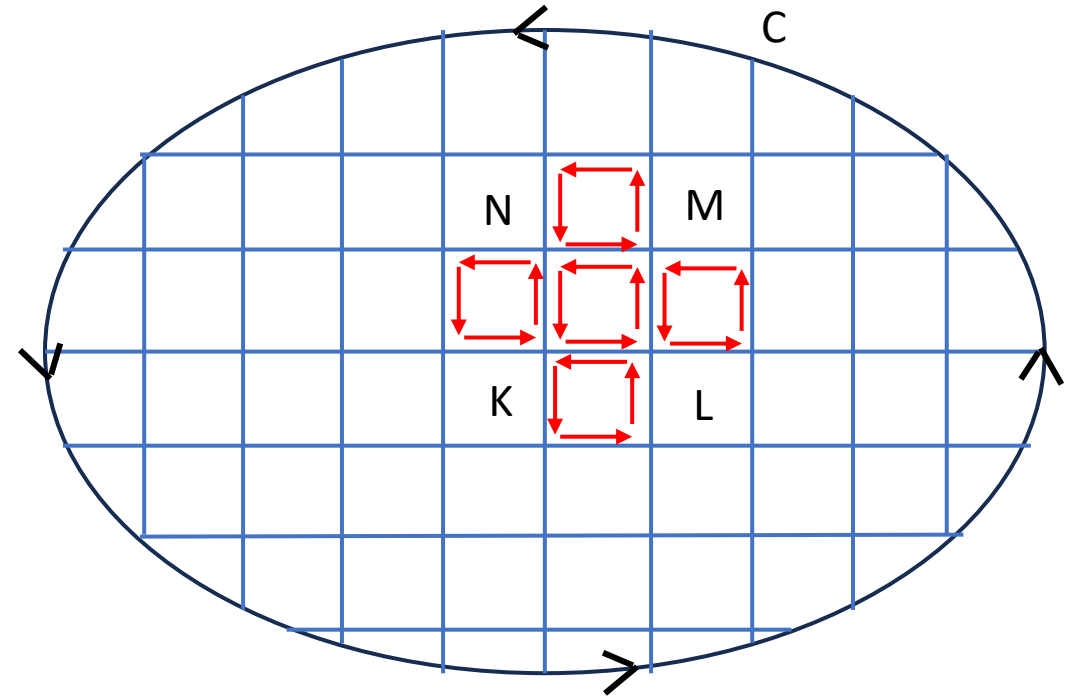
Stoke's Curl Theorem

- Suppose an area is enclosed by the closed path C in a vector field $\vec{A}$. This area can be considered to be made up of a large number of infinitesimal areas enclosed in infinitesimal closed paths (as shown in figure).
- Suppose in every area element, the circulation or curl of the vector field along the path is anticlockwise.
- Therefore the line integrals of the vector field at the common boundary of any two area elements will be mutually equal and opposite.
- For example

$$\int_K^L \vec{A} \cdot d\vec{l} = - \int_L^K \vec{A} \cdot d\vec{l}$$

$$\therefore \int_K^L \vec{A} \cdot d\vec{l} + \int_L^K \vec{A} \cdot d\vec{l} = 0$$

- Therefore, except for the outer boundary C, the sum of the line integrals along the common paths around all the area elements will be equal to zero. Hence the sum of line integrals along the boundaries of all the elements will be equal to the line integral of the vector field along the boundary C.



Stoke's Curl Theorem

$$\text{i.e.} \quad \oint_C \vec{A} \cdot d\vec{l} = \sum \oint_{dC} \vec{A} \cdot d\vec{l} \quad \dots(1)$$

According to the definition of curl,

$$\text{curl} \vec{A} \cdot \hat{n} = \lim_{da \rightarrow 0} \left[\frac{\oint_{dC} \vec{A} \cdot d\vec{l}}{da} \right]$$

$$\text{Or} \quad \oint_{dC} \vec{A} \cdot d\vec{l} = (\text{curl} \vec{A}) \cdot \hat{n} da \quad \dots(2)$$

Here $\hat{n}$ is the unit vector perpendicular to the $\vec{da}$ area element

$$\text{As } \hat{n} da = \vec{da}$$

Using equation (2) in equation (1)

$$\oint_C \vec{A} \cdot d\vec{l} = \sum (\text{curl} \vec{A}) \cdot \vec{da}$$

If the area $\vec{da}$ enclosed by the element dC tends to zero the summation be replaced by integration,

$$\oint_C \vec{A} \cdot d\vec{l} = \int_S (\text{curl} \vec{A}) \cdot \vec{da} \quad \dots(3)$$

Clearly, **the line integral of vector field along the limiting boundary of the area enclosed by a closed path is equal to the surface integral of the curl of the vector field. This is known as Stoke's Curl Theorem. This theorem is used in converting line integral into surface integral and vice versa.**

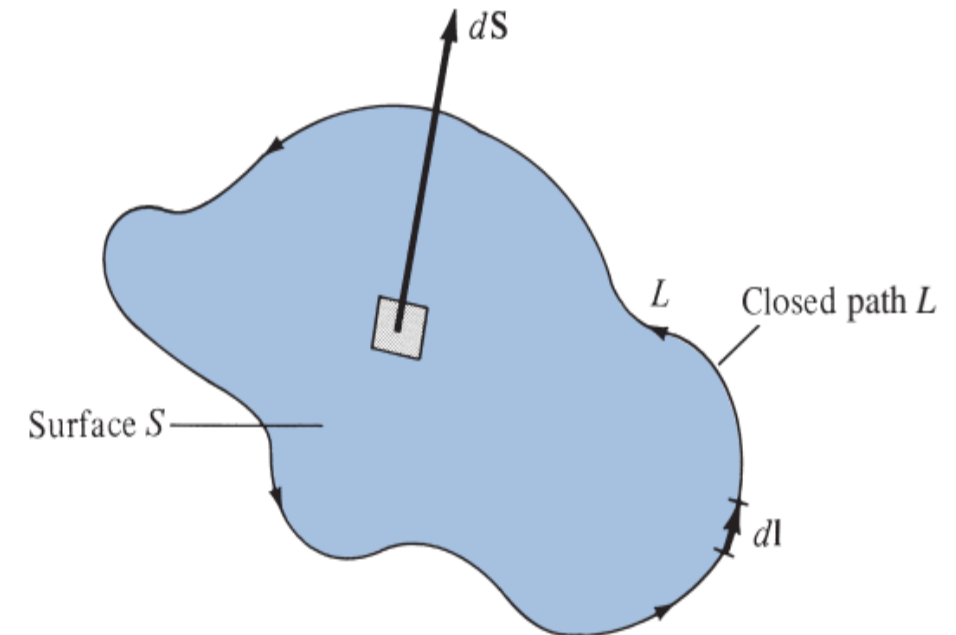
Stoke's Theorem

Also, from the definition of the curl of $\mathbf{A}$ in eq. (1), we may expect that

$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

This is called *Stokes's theorem*.

Stokes's theorem states that the circulation of a vector field $\mathbf{A}$ around a (closed) path L is equal to the surface integral of the curl of $\mathbf{A}$ over the open surface S bounded by), provided $\mathbf{A}$ and $\nabla \times \mathbf{A}$ are continuous on S .



Solid Angle

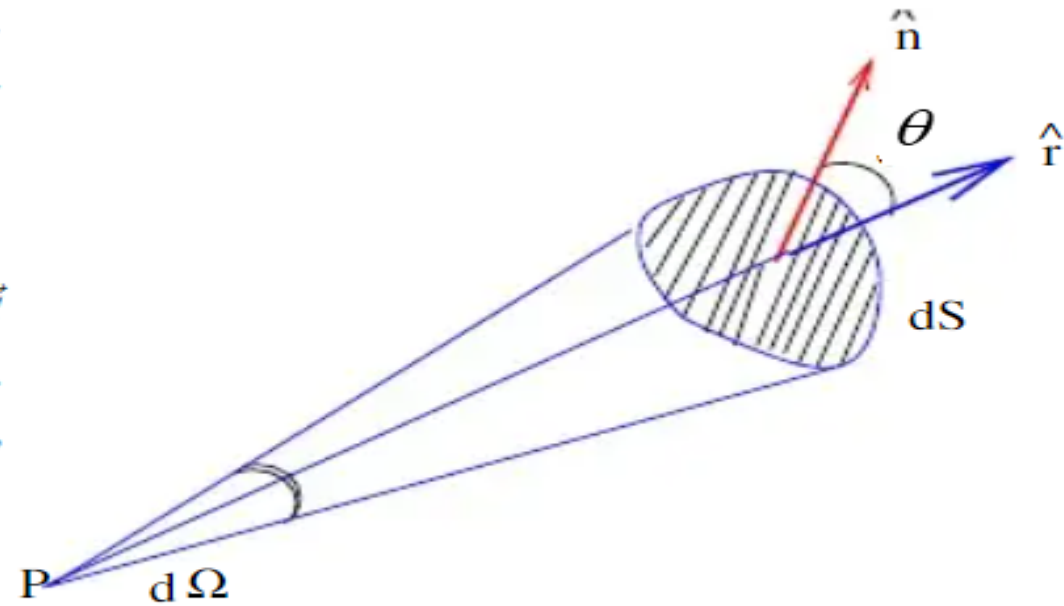
The concept of solid angle is a natural extension of a plane angle to three dimensions. Consider an area element dS at a distance r from a point P . Let $\hat{n}$ be the unit vector along the outward normal to dS .

The element of the solid angle subtended by the area element at P is defined as

$$d\Omega = \frac{dS_{\perp}}{r^2}$$

where $dS_{\perp}$ is the projection of $d\vec{S}$ along a direction perpendicular to $\vec{r}$. If θ is the angle between $\hat{r}$ and $\hat{n}$, then,

$$d\Omega = \frac{dS \cos \theta}{r^2}$$



Solid angle is dimensionless. However, for practical reasons it is measured in terms of a unit called **steradian** (much like the way a planar angle is measured in terms of degrees).

The maximum possible value of solid angle is 4π , which is the angle subtended by an area which encloses the point P completely.

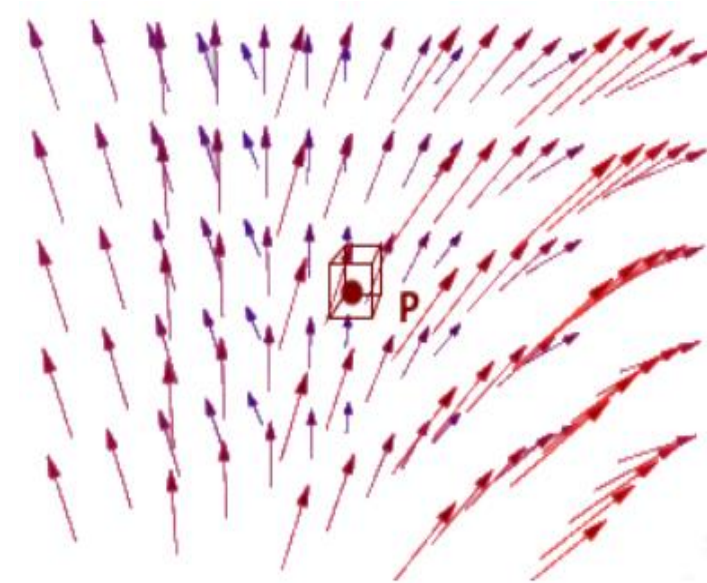
Divergence of Vector Field

- The net outflow of the flux of a vector field $\mathbf{A}$ from a closed surface S is obtained from the integral $\oint_S \vec{A} \cdot \vec{dS}$. We now define the divergence of $\mathbf{A}$ as the net outward flow of flux per unit volume over a closed incremental surface.
- The **divergence** of $\mathbf{A}$ at a given point P is the *outward* flux per unit volume as the volume shrinks about P .

Hence,

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v} \quad (\dots 1)$$

where Δv is the volume enclosed by the closed surface S in which P is located.



- Physically, we may regard the divergence of the vector field $\mathbf{A}$ at a given point as a measure of how much the field diverges or emanates or originates from that point.

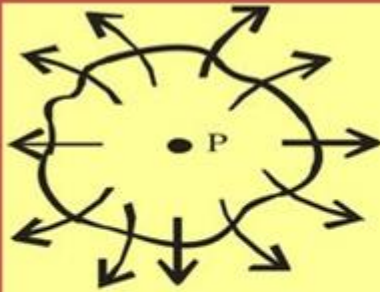
Summary of Divergence and Curl

(i) Divergence

➤ The divergence of a vector $\mathbf{V}$ written as $\text{div } \mathbf{V}$ represents the

$$\text{div } \mathbf{V} = \nabla \cdot \mathbf{V} =$$

$$\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$



(a) positive divergence



(b) negative divergence



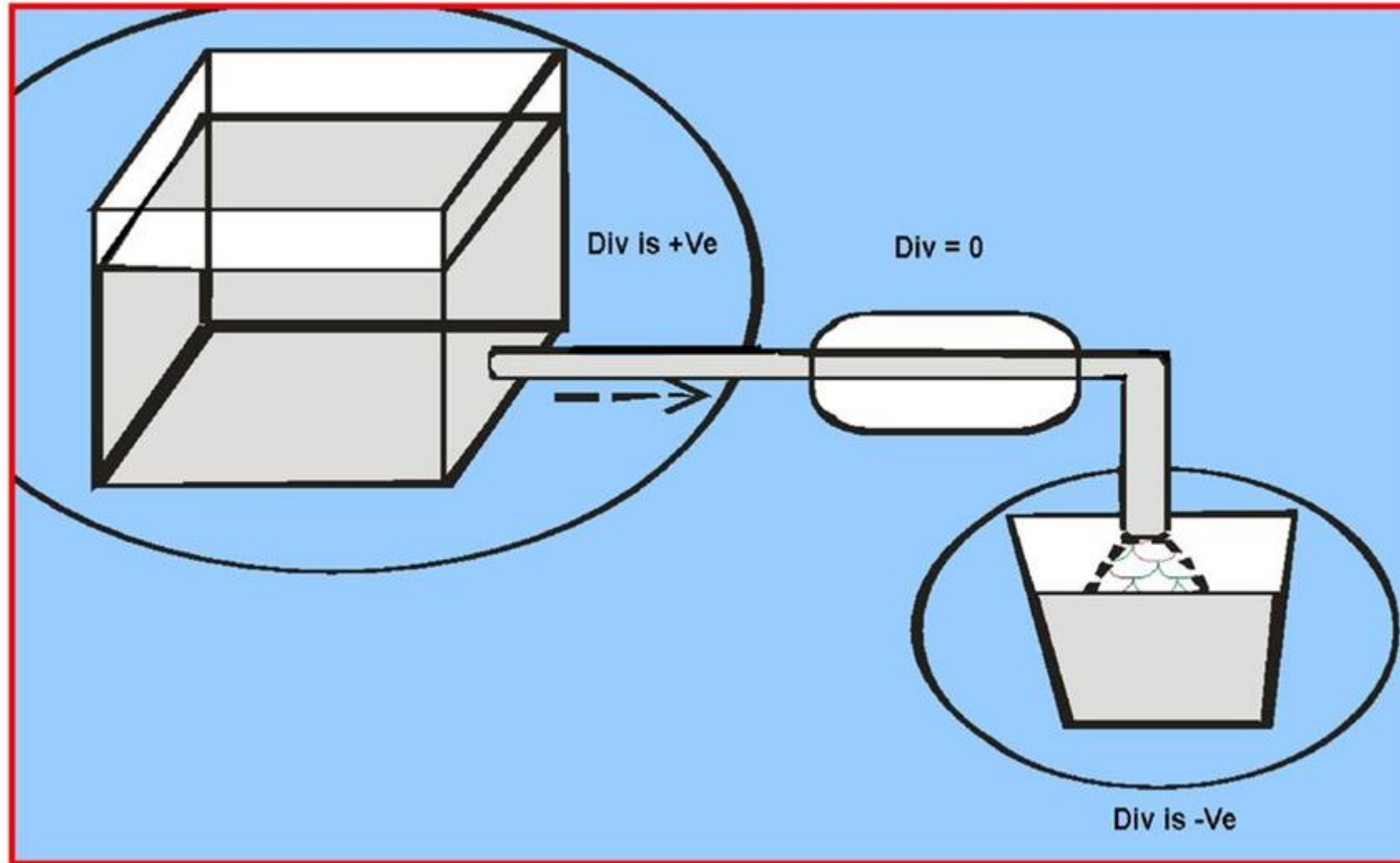
(c) zero divergence

Physical significance of divergence

- Physically the divergence of a vector quantity represents the rate of change of the field strength in the direction of the field.
- If the divergence of the vector field is positive at a point then something is diverging from a small volume surrounding with the point as a source.
- If it negative, then something is converging into the small volume surrounding that point is acting as sink.
- if the divergence at a point is zero then the rate at which something entering a small volume surrounding that point is equal to the rate at which it is leaving that volume.
- The vector field whose divergence is zero is called ***solenoidal***

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

Example for Divergence

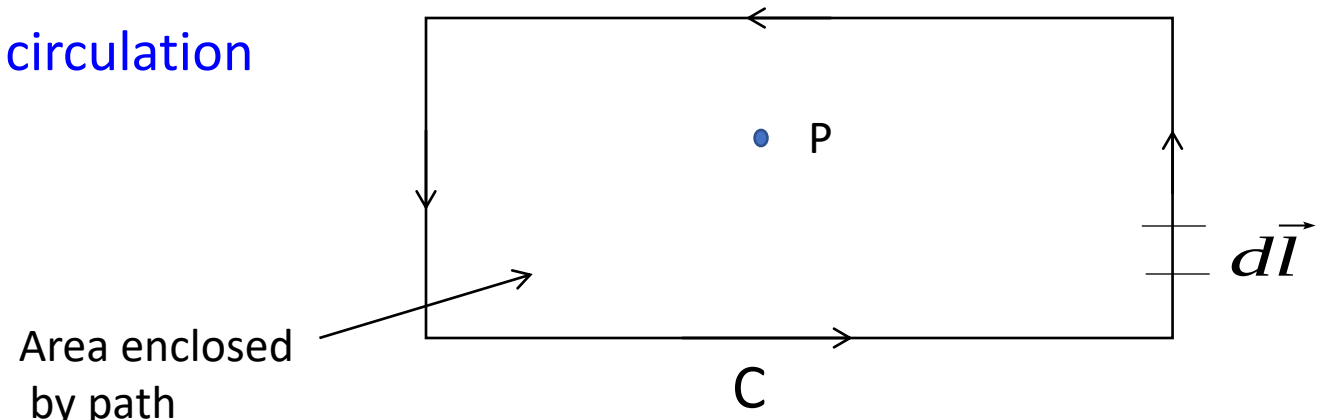


Curl of a Vector Field

- The line integral depends on the orientation of the surface enclosed by the closed path.
- If the surface enclosed by the closed path is kept in such a way that the line integral or circulation becomes maximum, then in this orientation the ratio of the line integral to the area enclosed by the path with the area tending to zero, is equal to the magnitude of the curl of the vector field at the position of the area element.

$$\text{curl} \vec{A} \cdot \hat{n} = \lim_{da \rightarrow 0} \left[\frac{\oint_C \vec{A} \cdot d\vec{l}}{da} \right]$$

- Its direction is perpendicular to the plane which contains the area element with maximum circulation around its boundary.



Curl of a Vector Field....

or

$$\text{curl} \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Physical significance of curl:

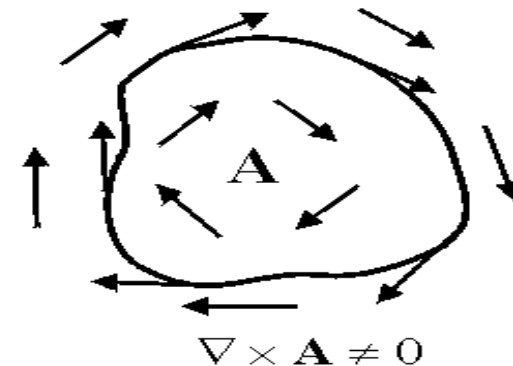
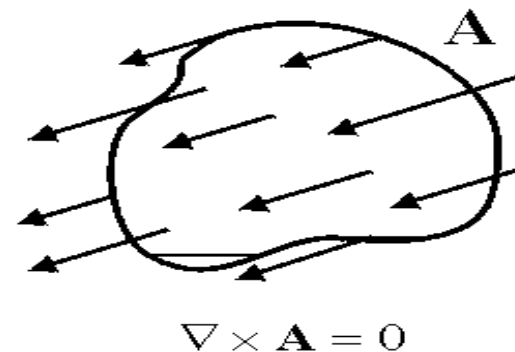
Curl of a vector field measure how much the vector $\vec{A}$ “curl around “ the point of observation. It comes from its rotational motion.

Example: For static field

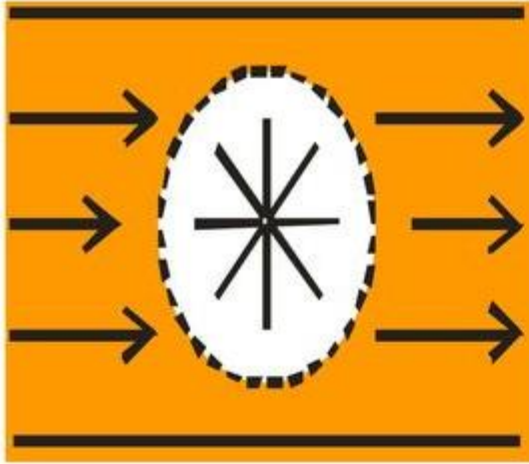
$$\nabla \times \vec{E} = 0 \quad (\text{Maxwell III equation})$$

$$\nabla \times \vec{B} = \mu_o J \quad (\text{Maxwell IV equation})$$

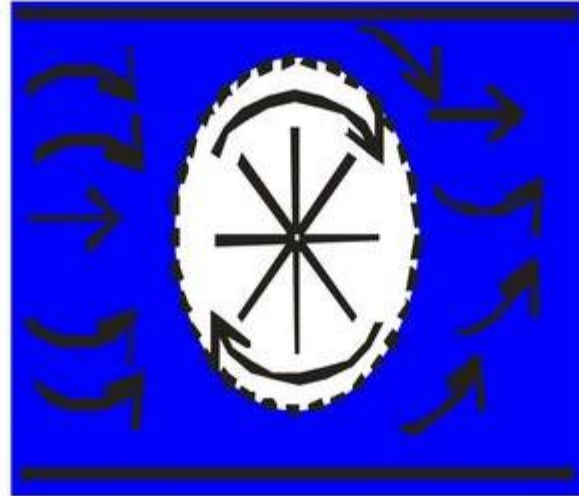
If curl of a vector field is zero, these vectors are called irrotational vector field.



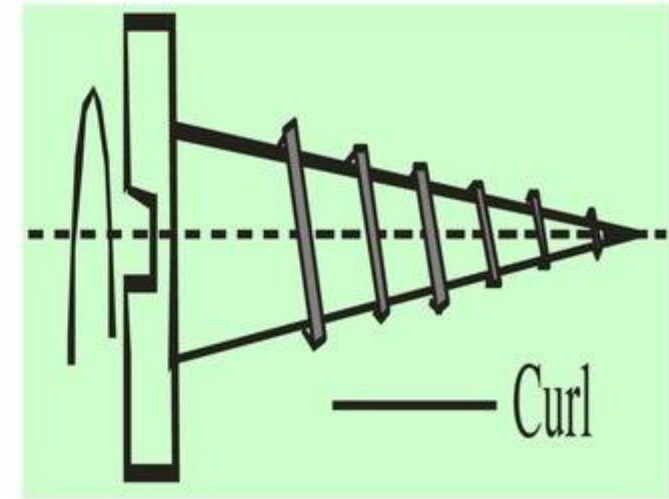
Representation of Curl



(I) No rotation of the paddle wheel represents zero curl



(II) Rotation of the paddle wheel showing the existence of curl



(III) direction of curl

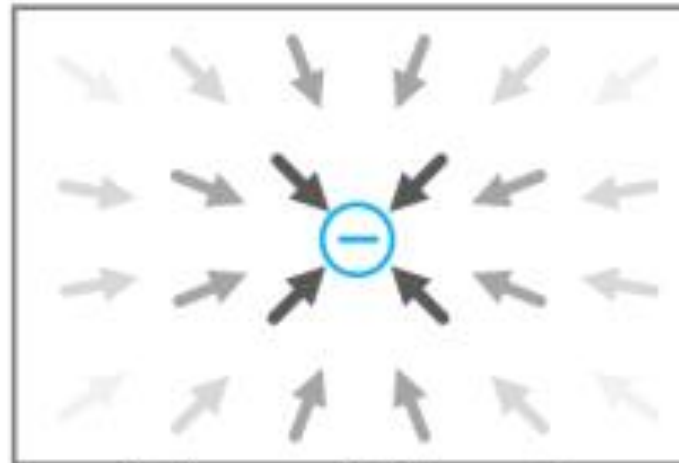
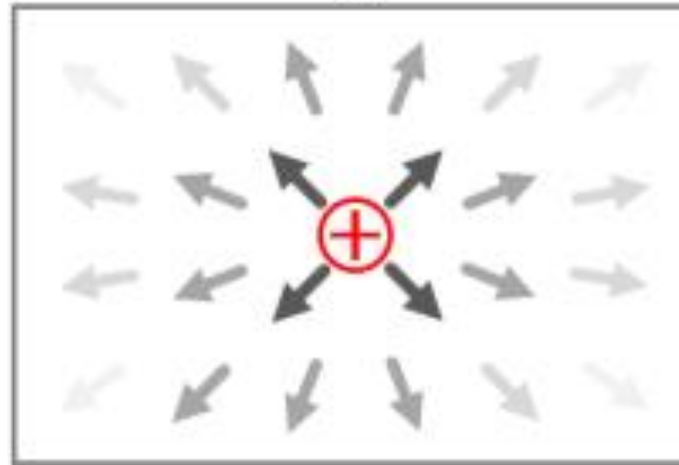
➤ For vector fields whose curl is zero there is no rotation of the paddle wheel when it is placed in the field, Such fields are called *irrotational*

Gradient



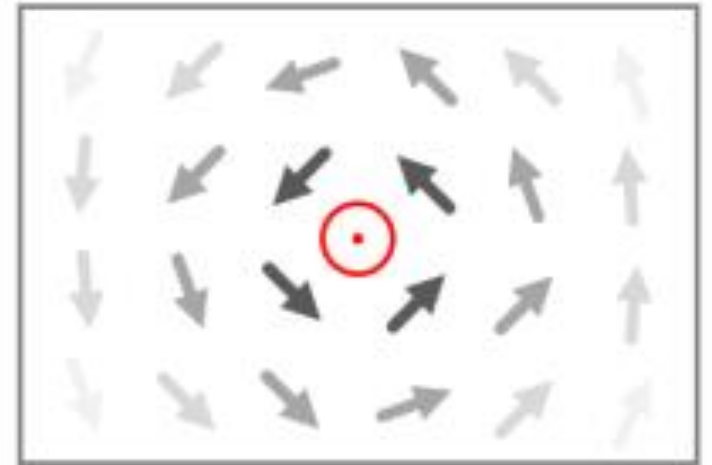
$\text{grad (scalar field)} = \text{vector}$

Divergence



$\text{div (vector field)} = \text{scalar}$

Curl



$\text{curl (vector field)} = \text{vector}$

Grad, Div, and the Curl

	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Conversion to Cartesian Coordinates		$x = \rho \cos \varphi \quad y = \rho \sin \varphi \quad z = z$	$x = r \cos \varphi \sin \theta \quad y = r \sin \varphi \sin \theta$ $z = r \cos \theta$
Vector A	$A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$	$A_\rho \hat{\rho} + A_\varphi \hat{\varphi} + A_z \hat{z}$	$A_r \hat{r} + A_\theta \hat{\theta} + A_\varphi \hat{\varphi}$
Gradient $\nabla \phi$	$\frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k}$	$\frac{\partial \phi}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial \phi}{\partial \varphi} \hat{\varphi} + \frac{\partial \phi}{\partial z} \hat{z}$	$\frac{\partial \phi}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial \phi}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial \phi}{\partial \varphi} \hat{\varphi}$
Divergence $\nabla \cdot A$	$\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$	$\frac{1}{\rho} \frac{\partial(\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$	$\frac{1}{r^2} \frac{\partial(r^2 A_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial A_\theta \sin \theta}{\partial \theta}$ $+ \frac{1}{r \sin \theta} \frac{\partial A_\varphi}{\partial \varphi}$
Curl $\nabla \times A$	$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$	$\begin{vmatrix} \frac{1}{\rho} \hat{\rho} & \hat{\varphi} & \frac{1}{\rho} \hat{z} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \varphi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\varphi & A_z \end{vmatrix}$	$\begin{vmatrix} \frac{1}{r^2 \sin \theta} \hat{r} & \frac{1}{r \sin \theta} \hat{\theta} & \frac{1}{r} \hat{\varphi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ A_r & r A_\theta & r A_\varphi \sin \theta \end{vmatrix}$

Gauss Divergence Theorem

- It relates the volume integral of the divergence of a vector $\mathbf{V}$ to the surface integral of the vector itself.
- According to this theorem, if a closed S bounds a volume τ , then

$$\int_{\tau} \text{div } \mathbf{V} \, d\tau = \mathbf{V} \bullet d\mathbf{s}$$

(or)

$$\int_{\tau} (\nabla \bullet \mathbf{V}) d\tau = \int_S \mathbf{V} \bullet d\mathbf{s}$$

Stoke's Theorem

- It relates the surface integral of the curl of a vector to the line integral of the vector itself.
- According to this theorem, for a closed path C bounds a surface S ,

$$\int_S (\text{curl } \mathbf{V}) \bullet d\mathbf{s} = \oint_C \mathbf{V} \bullet d\mathbf{l}$$

Gauss's Law

Q.3(b) State and define Gauss law in integral and differential form.

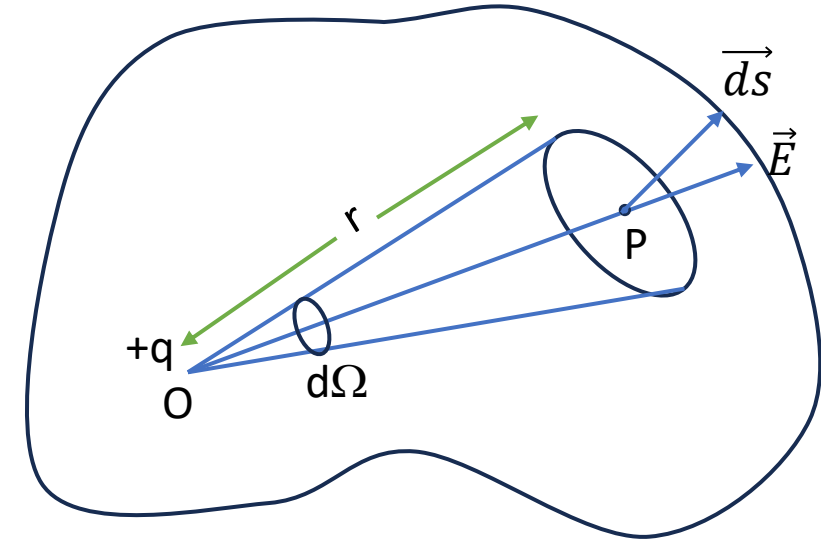
According to the Gauss's law for electrostatics, the total electric flux emerging from a closed surface situated in an electric field is equal to $\frac{1}{\epsilon}$ times the charge enclosed by it.

Suppose a point charge $+q$ placed at some point O is enclosed by an imaginary closed surface S (as shown in figure). The whole surface S can be divided into a number of small area elements. One of these area elements having area equal to $d\vec{s}$ is situated at P .

If the distance of the area element from the point charge $+q$ is r , then using the inverse square law, the intensity of electric field at the area element will be,

$$\vec{E} = \frac{1}{4\pi\epsilon} \frac{q}{r^2} \hat{r} \quad \dots(1)$$

Here $\hat{r}$ is the unit vector along OP and ϵ is the permittivity of the medium.



Gauss's Law

The electric flux emerging from the area element $\vec{ds}$ is given by

$$d\phi = \vec{E} \cdot \vec{ds}$$

or
$$d\phi = \frac{1}{4\pi\epsilon} \frac{q}{r^2} \hat{r} \cdot \vec{ds} \quad \dots(2)$$

If θ is the angle between $\hat{r}$ and $\vec{ds}$, then

$$\hat{r} \cdot \vec{ds} = |\hat{r}| ds \cos \theta = ds \cos \theta \quad (\text{from the concept of solid angle})$$

So that the electric flux
$$d\phi = \frac{1}{4\pi\epsilon} \frac{q ds \cos \theta}{r^2} \quad \dots(3)$$

From the definition of solid angle, the solid angle subtended at a point O by an area element ds at distance r is given by,

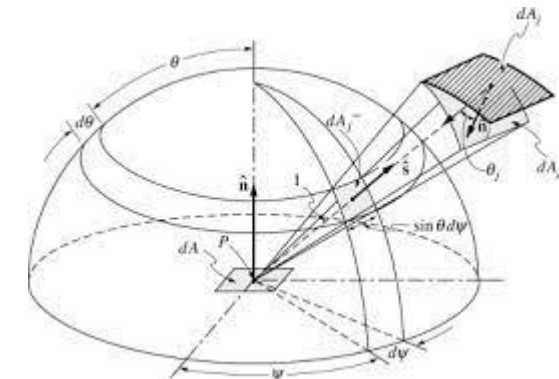
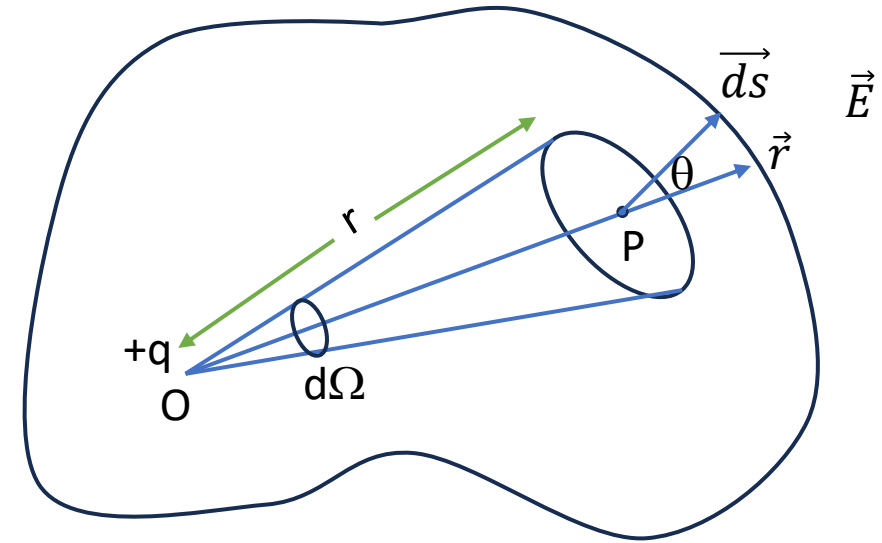
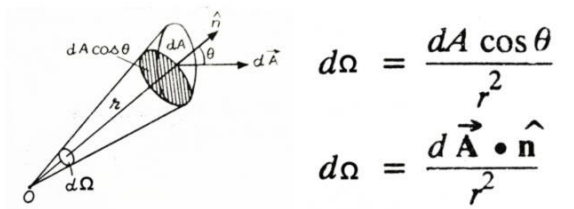
$$d\Omega = \frac{da}{r^2} = \frac{ds \cos \theta}{r^2}$$

$\therefore$ Electric flux emerging from the area element,

$$d\phi = \frac{q}{4\pi\epsilon} d\Omega \quad \dots(4)$$

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Solid angle (steradian (sr))



Gauss's Law....

Since the Electric flux emerging from the area element,

$$d\phi = \frac{q}{4\pi\epsilon} d\Omega \quad \dots(4)$$

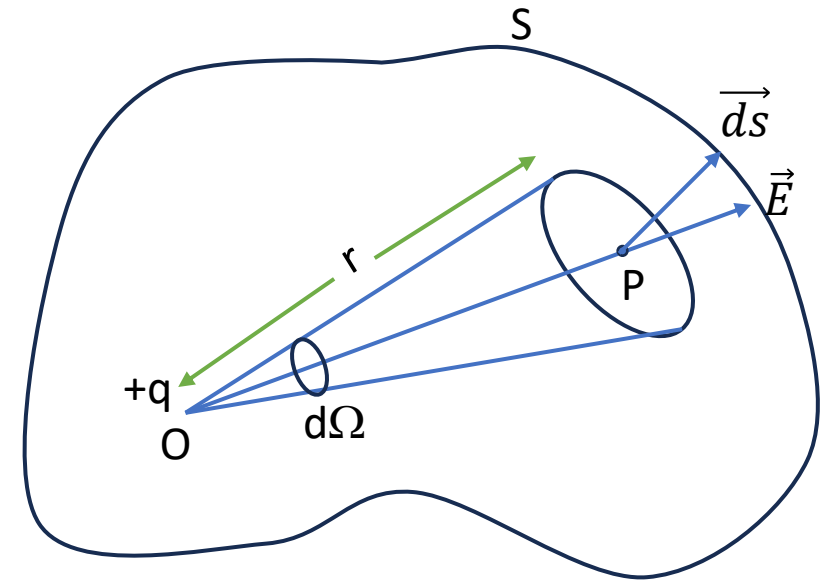
Electric flux is a scalar quantity, therefore the total flux emerging out from the closed surface will be equal to the sum of flux emerging from all the elements of the surface.

$\therefore$ Total flux emerging from the surface S,

$$\oint_S d\phi = \oint_S \frac{q}{4\pi\epsilon} d\Omega$$
$$\phi = \frac{q}{4\pi\epsilon} \oint_S d\Omega \quad \dots(5)$$

But the solid angle ($d\Omega$) formed by a closed surface at any point situated inside it is equal to 4π

$$\therefore \phi = \frac{q}{4\pi\epsilon} 4\pi$$
$$\phi = \frac{q}{\epsilon} \quad \dots(6)$$



Therefore, the total electric flux emerging from a closed surface situated in an electric field is equal to $\frac{1}{\epsilon}$ times the charge enclosed by it. This law is also applicable to charges distributed inside a closed surface.

Gauss's Law in Differential Form

Gauss's law expresses the relation between the electric flux emerging from an imaginary closed surface situated in an electric field and the charge existing inside this closed surface. **According to Gauss law, "the flux emerging out of a closed surface situated in an electric field is equal to the product of charge enclosed by the closed surface and $\frac{1}{\epsilon}$. Here ϵ is the permittivity of the medium".**

For vacuum, $\epsilon = \epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \frac{C^2}{N \times m^2}$ or Farad/m

Suppose several charges $q_1, q_2, q_3, \dots$ etc. are enclosed by an imaginary closed surface S .

$\therefore$ Total enclosed charge $q = q_1 + q_2 + q_3 + \dots$

If the intensity of electric field at the closed surface due to these charges is $\vec{E}$ then the total electric flux emerging from the closed surface S will be,

$$\phi = \oint_S \vec{E} \cdot \vec{da} \quad \dots(1)$$

According to Gauss's law

$$\phi = \oint_S \vec{E} \cdot \vec{da} = \frac{q}{\epsilon} \quad \dots(2)$$

This is integral form of Gauss's Law

Gauss's Law in Differential Form

$$\phi = \oint_S \vec{E} \cdot \vec{da} = \frac{q}{\epsilon} \quad \dots(2)$$

If the charge enclosed by the closed surface is distributed in the enclosed volume and ρ is the charge density, then

$$\text{charge density} = \frac{\text{charge}}{\text{volume}}$$

or
$$\rho = \frac{dq}{dV} \quad \dots(3)$$

Here dq is the amount of charge in the volume element dV .

$$\therefore dq = \rho dV$$

$\therefore$ Total amount of charge in the volume enclosed by the complete surface S ,

$$q = \oint_S dq = \oint_V \rho dV \quad \dots(4)$$

Substituting equation (4) in equation (2),

$$\phi = \oint_S \vec{E} \cdot \vec{da} = \frac{1}{\epsilon} \oint_V \rho dV \quad \dots(5)$$

Gauss's Law in Differential Form

$$\phi = \oint_S \vec{E} \cdot \vec{da} = \frac{1}{\epsilon} \oint_V \rho dV \quad \dots(5)$$

Using **Gauss divergence theorem** for changing the surface integral of the electric field into volume integral,

$$\int_S \vec{E} \cdot \vec{da} = \int_V (\nabla \cdot \vec{E}) dV \quad \dots(6)$$

Comparing equations (5) and (6),

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon} \quad \dots(7)$$

The above equation is the differential form of Gauss's law.

Applications of Gauss's Law

Gauss law can be used to find electric field intensity at some distance due to different charge distribution.

Step to determine Electric field intensity:

- Draw a suitable Gaussian surface enclosing the point where we want to calculate electric field intensity.

Gaussian Surface:

An imaginary closed surface of any shape drawn in an electric field to solve problems concerning electric flux is called Gaussian surface. The shape of the Gaussian surface is chosen on the basis of symmetry of the problem, so that the expression for Gauss's law

$$\phi = \oint_S \vec{E} \cdot \vec{da} \quad \text{can be evaluated conveniently.}$$

- Calculate total electric flux using above formula.

- Calculate electric flux using equation $\phi = \frac{q}{\epsilon}$

- Compare the two equations to get electric field intensity

Applications of Gauss's Law

- Deduction of Coulomb's Law (Point Charge)
- Electric field due to a charged solid sphere (Uniformly Charged Sphere)
- Field due to line of charge
- Field due to uniformly charged plane (sheet of charge)
- Field due to infinite charged conducting plate
- Field due to two parallel charged plates.
- Field due to a spherically symmetric charge distribution (Volume distribution of charge)

Applications of Gauss's Law

Deduction of Coulomb's Law (Point Charge)

Let us consider electric field due to a single isolated positive charge of magnitude q . Field of a positive charge is in radially outward direction everywhere and magnitude of electric field intensity is same for all points at a distance r from the charge. We can assume Gaussian surface to be sphere of radius r enclosing the charge q . From Gauss's law

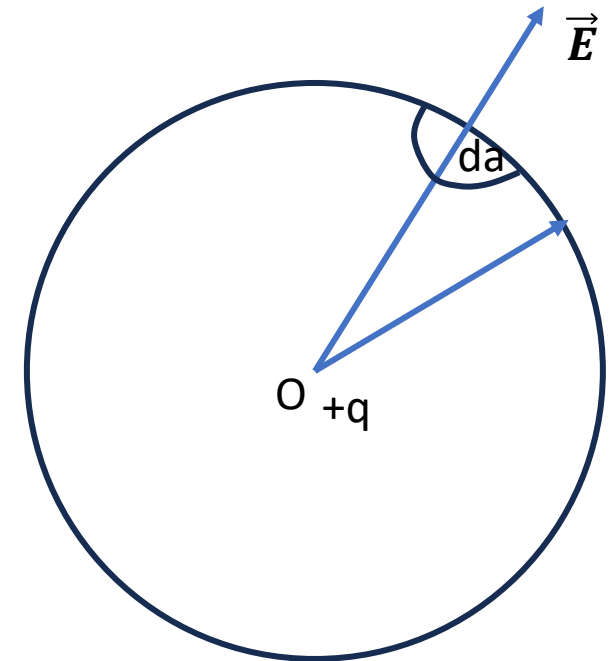
$$\oint \vec{E} \cdot \vec{da} = E \oint \vec{da} = \frac{q}{\epsilon_0}$$

Since E is constant at all points on the surface therefore,

$$EA = \frac{q}{\epsilon_0} \quad \text{or} \quad E = \frac{q}{\epsilon_0 A}$$

Since surface area of the sphere is $A = 4\pi r^2$, thus

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$



Applications of Gauss's Law

Now the force acting on a point charge q' at a distance r from point charge q is

$$F = q'E$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{qq'}{r^2}$$

This is nothing but the mathematical statement of Coulomb's law



Applications of Gauss's Law

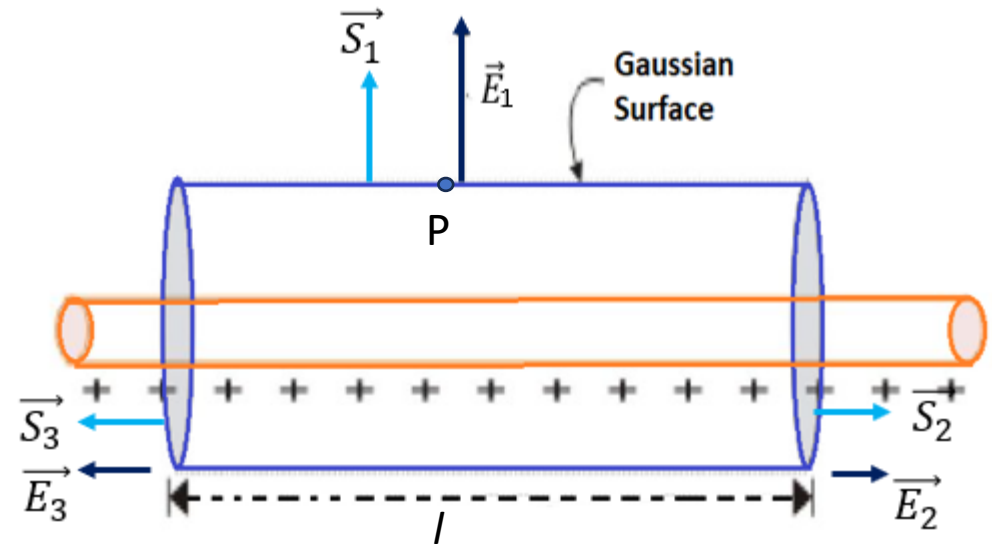
Electric Field due to line charge (Infinite Line Charge)

Let us consider a section of an infinite rod of charge with the linear charge density λ (charge per unit length) constant for all points on the line. Let the line charge is lying parallel to x- axis. Consider a point P at a distance r from the linear charge at which electric field strength has to be calculated. We choose a circular cylinder of radius r and length l as a Gaussian surface, closed at each end normal to the axis. The Gaussian surface can be divided into three sections S_1 where P is lying and S_2 and S_3 , the surface of the ends of Gaussian cylinder. The total charge enclosed by the Gaussian surface is $q = \lambda l$.

Now applying Gauss's law, we get

$$\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

$$\text{or} \quad \left[\int_{S_1} \vec{E}_1 \cdot d\vec{s}_1 + \int_{S_2} \vec{E}_2 \cdot d\vec{s}_2 + \int_{S_3} \vec{E}_3 \cdot d\vec{s}_3 \right] = \frac{q}{\epsilon_0}$$



Cylindrical Gaussian Surface for calculation of electric field due to line charge

Applications of Gauss's Law

Electric Field due to line charge

Because of the symmetry $\vec{E}_2 = \vec{E}_3 = 0$

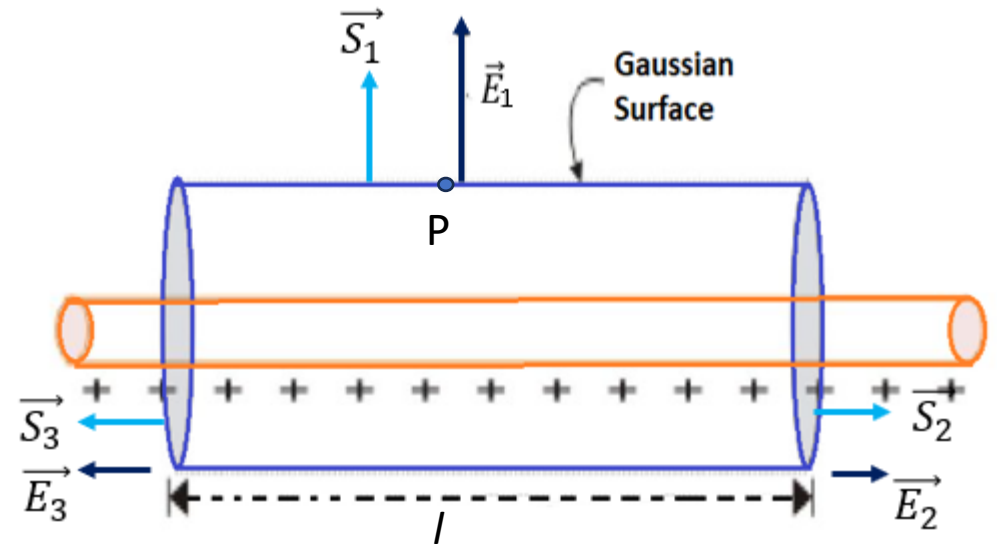
And let $\vec{E}_1 = \vec{E}$, then

$$\int_{S_1} \vec{E} \cdot d\vec{s}_1 = \frac{q}{\epsilon_0}$$

Or
$$\int_{S_1} \vec{E} \cdot d\vec{s}_1 = \frac{\lambda l}{\epsilon_0}$$

Since E is constant at the surface and $\vec{E}$ and S_1 are parallel everywhere, we have

$$\int_{S_1} \vec{E} \cdot d\vec{s}_1 = E \int_{S_1} d\vec{s}_1 = \frac{\lambda l}{\epsilon_0}$$



Cylindrical Gaussian Surface for calculation of electric field due to line charge

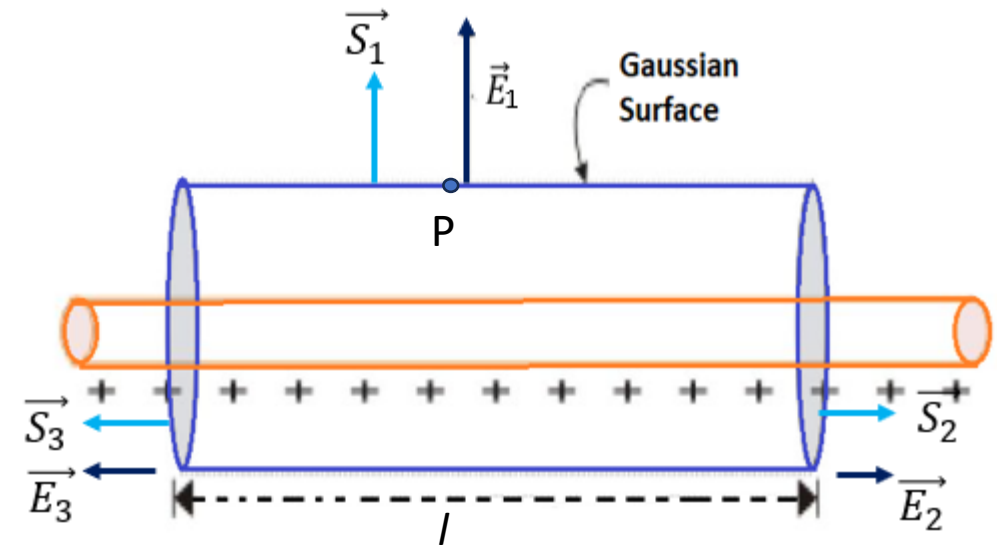
Applications of Gauss's Law

But $\int_{S_1} \vec{ds}_1 = 2\pi r l$

$\therefore E 2\pi r l = \frac{\lambda l}{\epsilon_0}$

Or $E = \frac{\lambda}{2\epsilon_0 \pi r}$

Which is radially directed along the cylinder.



Cylindrical Gaussian Surface for calculation of electric field due to line charge

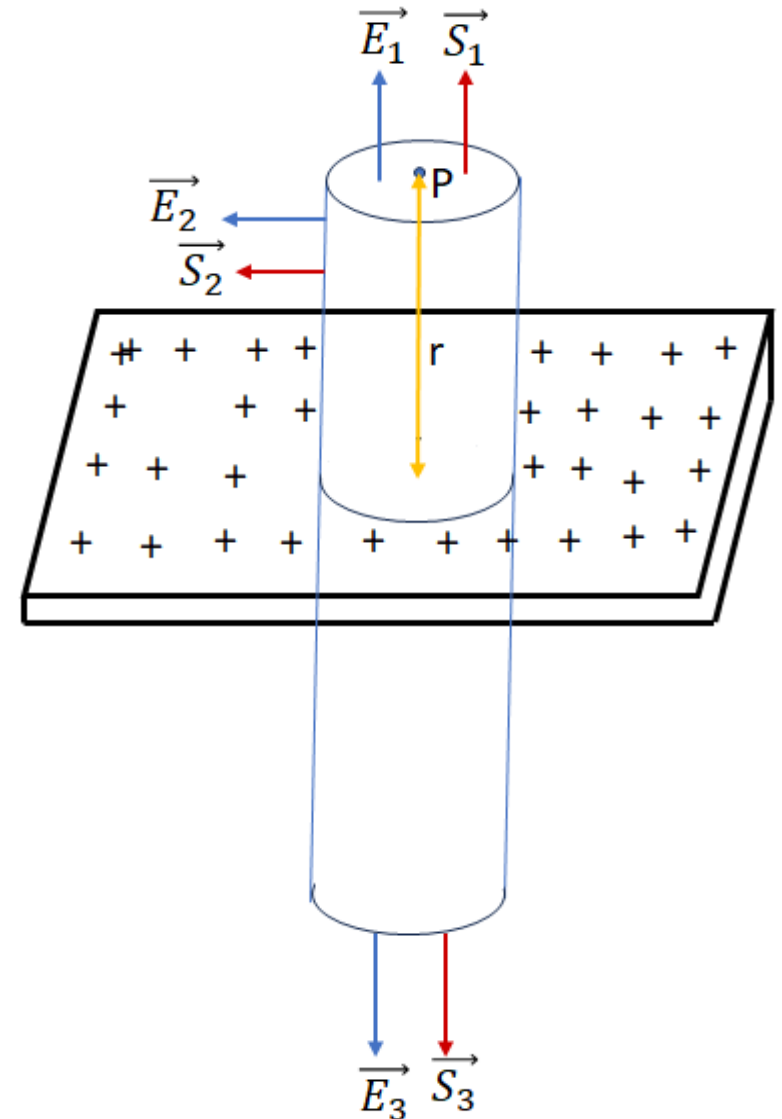
Applications of Gauss's Law

Electric Field due to a sheet of charge (Infinite Sheet of Charge)

Let us consider a portion of thin, non-conducting, infinite sheet of charge with surface charge density σ (charge per unit area) being constant. Consider a point P at a distance r from the surface of the sheet. Now draw a cylindrical surface having its upper end through P and lower end at a distance r from the sheet. Let S_1 , S_3 be the surface areas of the ends of the cylinder and S_2 the surface area of the cylindrical surface. From symmetry, E points at right angles to the ends of the cylinder and away from the plane. Since E does not pierce the cylindrical surface, there is no contribution to the flux from this surface. Thus, according to Gauss's law, we have the

$$\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} = \frac{\sigma S}{\epsilon_0}$$

$$\text{Or } \left[\int_{S_1} \vec{E}_1 \cdot d\vec{s}_1 + \int_{S_2} \vec{E}_2 \cdot d\vec{s}_2 + \int_{S_3} \vec{E}_3 \cdot d\vec{s}_3 \right] = \frac{\sigma S}{\epsilon_0}$$



Applications of Gauss's Law

Since $\vec{E}_2 = 0$, $\vec{E}_1 = -\vec{E}_3 = \vec{E}$ (say)

And $d\vec{s}_1 = -d\vec{s}_3 = d\vec{s}$ (say)

Also $\vec{E}_1 \parallel d\vec{s}_1$, $\vec{E}_3 \parallel d\vec{s}_3$

We have

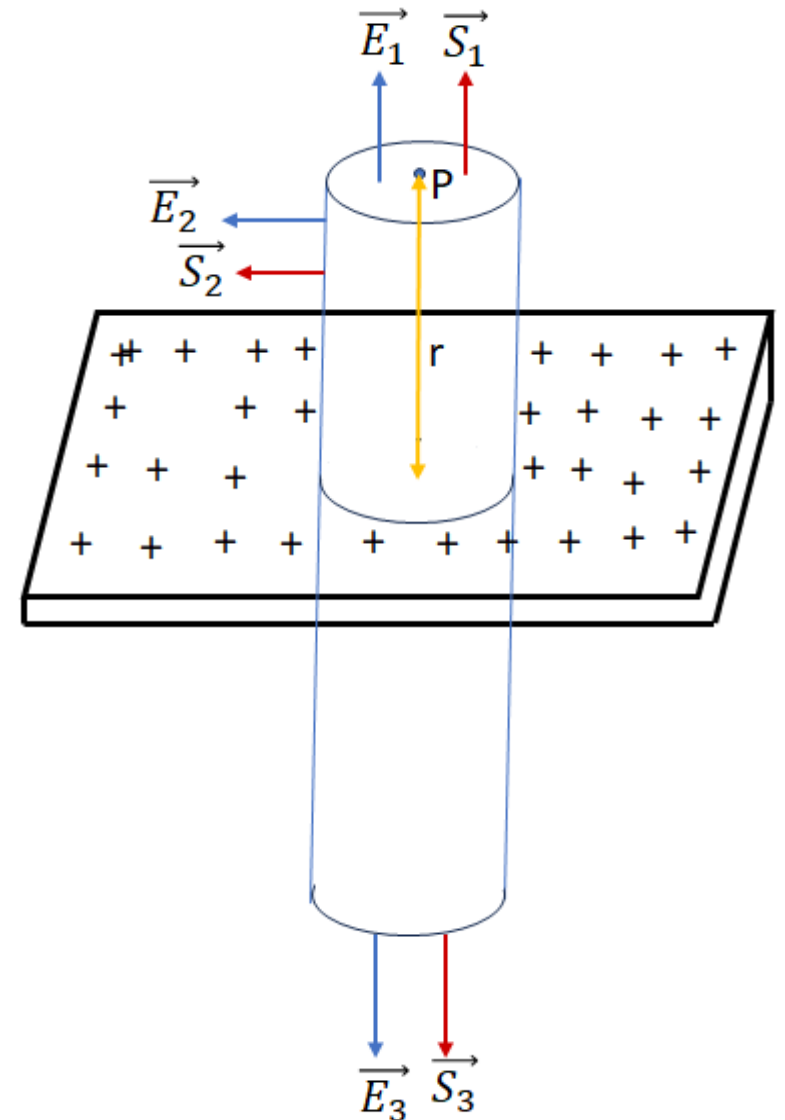
$$\int_{S_2} \vec{E}_2 \cdot d\vec{s}_2 = 0$$

$$\int_{S_1} \vec{E}_1 \cdot d\vec{s}_1 = \int \vec{E} \cdot d\vec{s} = E \int d\vec{s} = ES$$

$$\text{And } \int_{S_3} \vec{E}_3 \cdot d\vec{s}_3 = \int (-\vec{E}) \cdot (d\vec{s}) = E \int d\vec{s} = ES$$

$$\therefore ES = \frac{\sigma S}{2\epsilon_0} \quad \text{Or} \quad E = \frac{\sigma}{2\epsilon_0}$$

Thus, E is same for all points on each side of the plane.



Applications of Gauss's Law

Electric Field due to Infinite Parallel Sheets of Charge

Let P_1 and P_2 be two infinite plane parallel sheets carrying opposite uniform charge densities. Let a point P be lying in the region between two charged plates. Now in order to calculate the electric field at point P , draw a normal cylinder of cross-sectional area S through point P having one of its ends in the plate. Using Gauss's law, we have

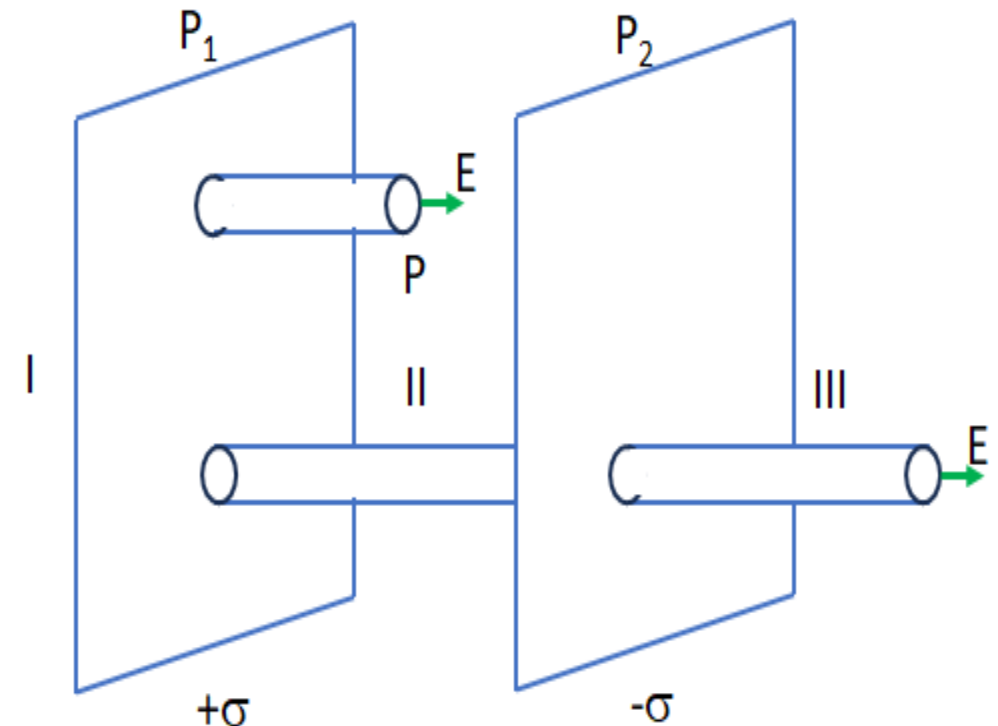
$$\oint_S \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0}$$

Let σ be the surface charge density, then $q = \sigma S$

$\therefore$ Above equation becomes $ES = \frac{\sigma S}{\epsilon_0}$ [As $\vec{E}$ and $\vec{S}$ are parallel]

$$\text{or } E = \frac{\sigma}{\epsilon_0}$$

which implies the the field between two plates is uniform and independent of the separation between them.



Applications of Gauss's Law

Again consider a point Q lying extreme right of the two plates. Now draw a cylinder of cross-sectional area S through this point Q with one of its face lying on plate P_1 . Therefore from Gauss's law, we have

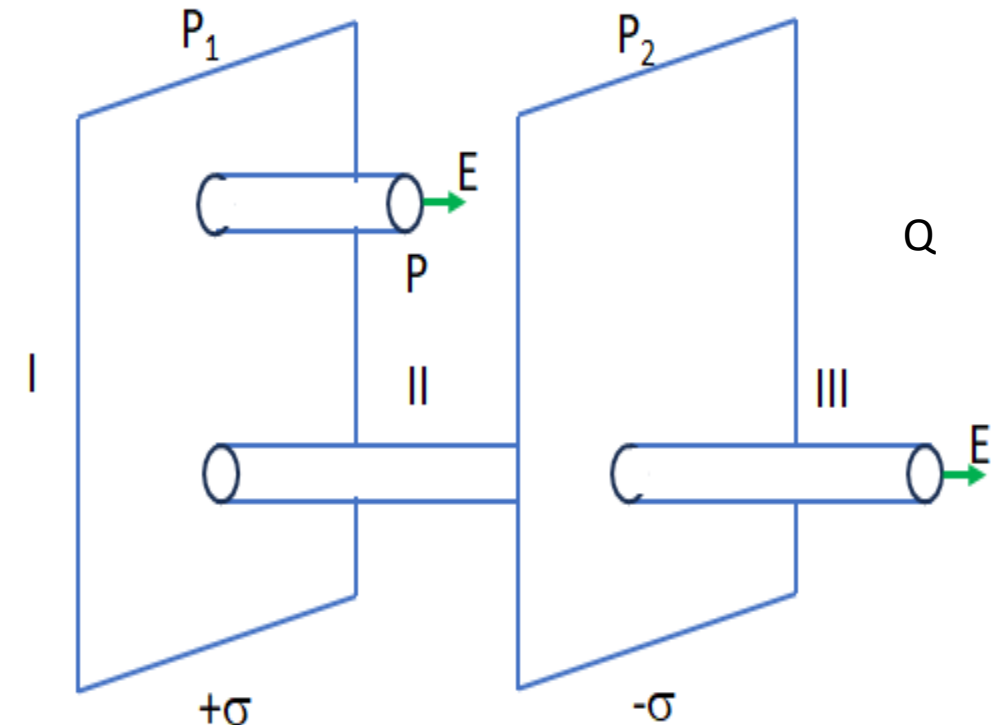
$$\oint_S \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0}$$

Since the charge inside the Gaussian surface is zero i.e. $q = \sigma S - \sigma S$, therefore we have

$$\oint_S \vec{E} \cdot \vec{ds} = 0$$

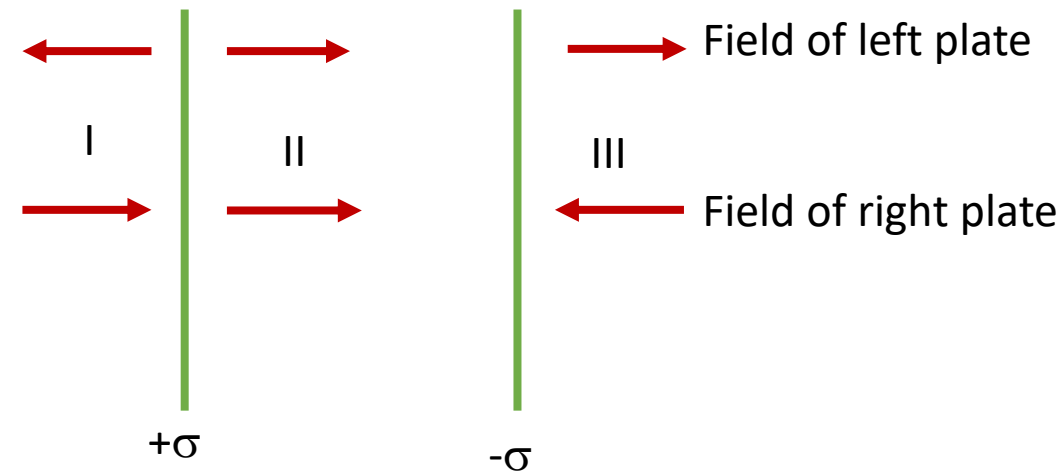
Or $ES=0 \Rightarrow E=0$

which clearly indicates that the electric field intensity due to two oppositely charged infinite plane sheets at a point outside the plates is zero.



Applications of Gauss's Law

We can also verify these results pictorially as follows:
We know that the electric field intensity at any point at a distance r due to an infinite plane sheet of charge having charge density σ is $\frac{\sigma}{2\epsilon_0}$. Therefore the left plate produces a field $\frac{\sigma}{2\epsilon_0}$ which point away from it (as shown in the figure) to the left in region I and to the right in region II and III. The right plate produces a field $\frac{\sigma}{2\epsilon_0}$ which points towards it to the right in regions I and II and to the left in region III. The two fields cancel in region I and II but they conspire in region III. Thus, between the plates the electric field intensity is $\frac{\sigma}{\epsilon_0}$ and outside is zero.



Applications of Gauss's Law

Q.4(a) Find the electric field at a point P outside a uniformly charged sphere.

Q.2(b) Apply Gauss law to find out the electric field due to a uniformly charged solid sphere at a point [5] outside and inside the sphere.

The electric field due to the Spherical Charge Distribution (Uniformly Charged Sphere)

Consider a sphere of charge Q having volume density ρ . Let r be the radius of sphere. Let P and P' be two observation points outside and inside the sphere respectively as shown in figure.

To calculate the electric field strength at P and P' , we consider Gaussian surfaces S_1 and S_2 respectively through points P and P' with O as their center. Let us consider $OP = r_o$ and $OP' = r_i$.

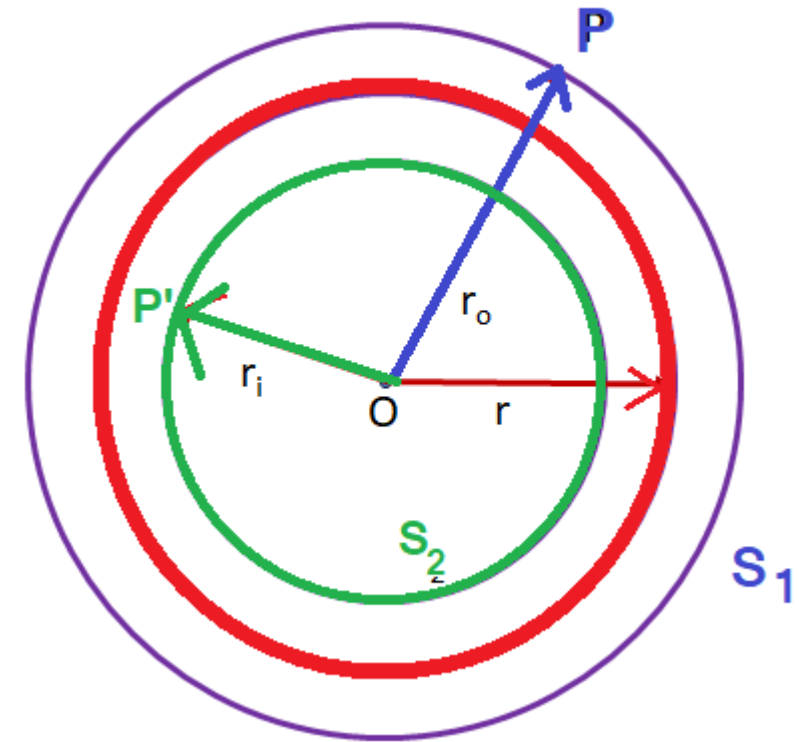
Now apply Gauss's law to the outer Gaussian surface (S_2) through P we have,

$$\oint_S \vec{E}_0 \cdot d\vec{s}_1 = \frac{Q}{\epsilon_0}$$

Where $\vec{E}_0$ is electric field at point P and charge $Q = \frac{4}{3}\pi r^3 \rho$

Or

$$\epsilon_0 \oint_S \vec{E}_0 \cdot d\vec{s}_1 = Q$$



A spherically symmetric charge distribution, showing two Gaussian surfaces

Applications of Gauss's Law

- Spherical Charge Distribution

$$\epsilon_0 E_0 \oint_S \vec{ds}_1 = Q$$

$$\epsilon_0 E_0 4\pi r_o^2 = Q$$

(The surface area of sphere is $4\pi r^2$)

or

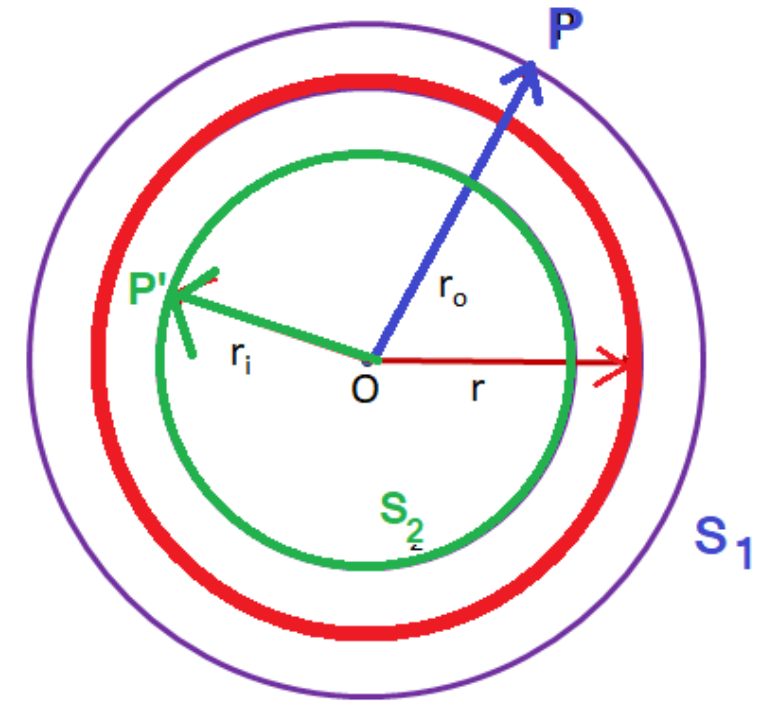
$$\vec{E}_0 = \frac{Q}{4\epsilon_0\pi r_o^2}$$

Thus, it is clear that the sphere of charge behaves as if the whole charge is concentrated at the center. Again apply Gauss's law to the inner Gaussian surface (S_2) through P' , we have

$$\oint_S \vec{E}_i \cdot \vec{ds}_2 = \frac{Q'}{\epsilon_0}$$

where $\vec{E}_i$ is the electric field at P' and Q' is the charge enclosed by the Gaussian surface S_2 . Therefore $Q' = \frac{4}{3}\pi r_i^3 \rho$

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A spherically symmetric charge distribution, showing two Gaussian surfaces

Applications of Gauss's Law

- Spherical Charge Distribution

$$\text{Or } Q' = \frac{4}{3}\pi r_i^3 \rho \frac{Q}{\frac{4}{3}\pi r^3} = Q \frac{r_i^3}{r^3}$$

$$\therefore \oint_S \epsilon_0 \vec{E}_i \cdot \vec{ds}_2 = Q' = Q \frac{r_i^3}{r^3}$$

$$\epsilon_0 E_i \oint_S \vec{ds}_2 = Q \frac{r_i^3}{r^3}$$

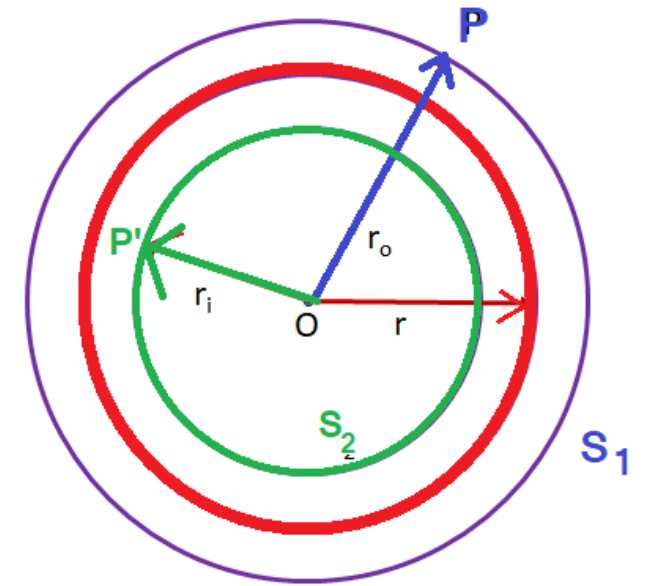
$$\epsilon_0 E_0 4\pi r_i^2 = Q \frac{r_i^3}{r^3}$$

$$E = \frac{Q}{4\pi\epsilon_0} \frac{r_i}{r^3}$$

The above relation shown that the intensity of the electric field inside the sphere of charge varies directly as a distance from the center.

Suppose $r_i = r$, then the electric field due to the sphere of the charge Q will be $\Rightarrow E = \frac{Q}{4\pi\epsilon_0} \frac{1}{r^2}$

which represents the electric field due to the sphere of radius r just on the surface of the sphere.



A spherically symmetric charge distribution, showing two Gaussian surfaces

BIT End Sem Question :- Show that the electrostatic field E is conservative in nature

Q.3(a) Show that electrostatic field is conservative in nature.

Suppose a charge Q is situated in an electric field at points A and B. The path from A to B is a closed path. The electric field does work to move a test charge from point A to B. To determine the work done, we sum up the work done for all the small segments that the path A to B is divided into. This kind of integral evaluated along a line is known as a line integral.

The line integral of the electric field along A to B is as follows:

$$\int_A^B \vec{E} \cdot d\vec{l}$$

The line integral of the electric field along B to A is as follows:

$$\int_B^A \vec{E} \cdot d\vec{l}$$

Therefore, the line integral along the closed path as depicted in the figure above is:

$$\int_A^B \vec{E} \cdot d\vec{l} + \int_B^A \vec{E} \cdot d\vec{l}$$

However,

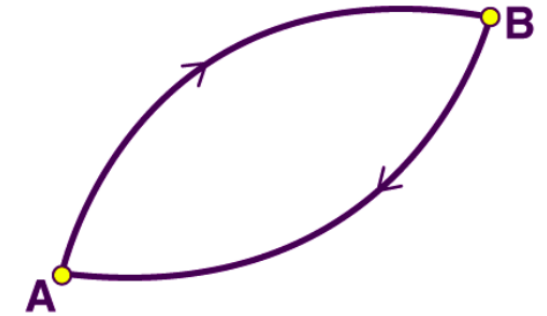
$$\int_B^A \vec{E} \cdot d\vec{l} = - \int_A^B \vec{E} \cdot d\vec{l}$$

$$\int_A^B \vec{E} \cdot d\vec{l} + \int_B^A \vec{E} \cdot d\vec{l} = \int_A^B \vec{E} \cdot d\vec{l} + - \int_A^B \vec{E} \cdot d\vec{l}$$

This demonstrates that the line integral of an electric field along a closed path is zero.

Therefore, the work done by the electric field is independent of the path, meaning it relies only on points A and B.

This confirms that the electric field is indeed conservative in nature.



Q4 (b) Prove the relation $D = \epsilon_0 E + P$.

The relationship among the three vectors D , E , P in the metre-kilogram-second (mks) or SI system is: $D = \epsilon_0 E + P$
(ϵ_0 is a constant, the permittivity of a vacuum)

Curl of the Electric Field:

Let's take the simplest electric field:

Electric field due to a single point charge q is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

We need to find the curl of it

$$\nabla \times \mathbf{E}(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \nabla \times \left(\frac{1}{r^2} \hat{\mathbf{r}} \right)$$

Take the area integral

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \int_{surf} \nabla \times \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \cdot d\mathbf{a}$$

Use Stokes's theorem

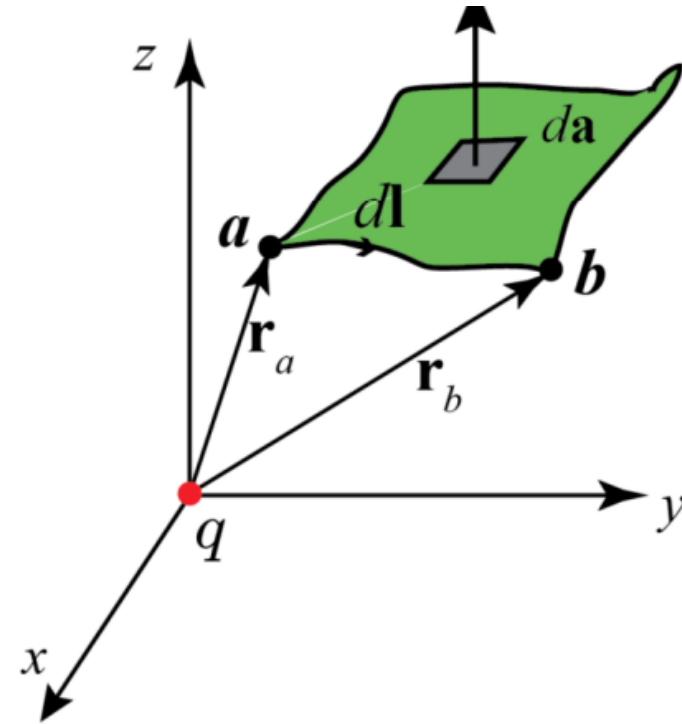
$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \oint \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \cdot d\mathbf{l} = \frac{q}{4\pi\epsilon_0} \oint \frac{1}{r^2} dr$$

since

$$d\mathbf{l} = dr\hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin\theta d\phi \hat{\boldsymbol{\phi}}$$

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \times \frac{1}{r} \Big|_{r_a}^{r_a} = 0 \text{ implies}$$

$$\nabla \times \mathbf{E} = 0$$



Curl of an electric field ($\nabla \times \mathbf{E}$) is zero . We have shown this for the simplest field, which is the field of a point charge. But it can be shown to be true for any electric field, as long as the field is static.

if the field is dynamic, that is, what if the field changes as a function of time?

Ans:- $\nabla \times \mathbf{E} = - d\mathbf{B}/ dt$

Gauss's Law from Coulomb's Law:

If Coulomb's Law and Gauss's law have the same information content, can we derive Gauss's law from Coulomb's law?

Coulomb's law gives the electric field due to a volume charge $\rho(\mathbf{r}')$

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{\mathbf{r}}}{r^2} \rho(\mathbf{r}') d\tau$$

Take the divergence of both sides of the equation

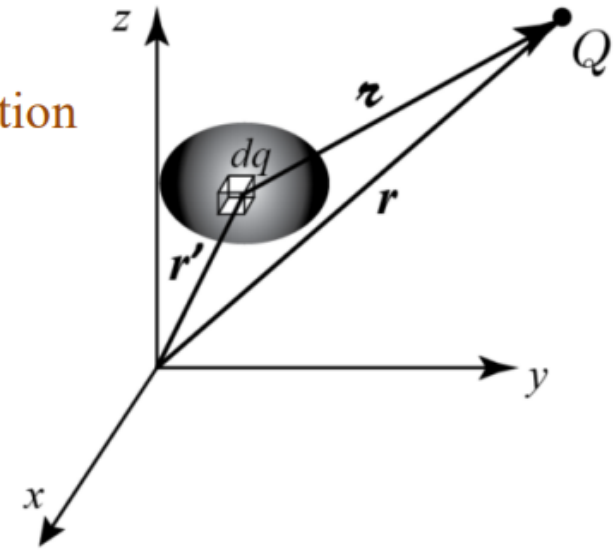
$$\nabla \cdot \mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) \rho(\mathbf{r}') d\tau$$

We have: $\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta(\mathbf{r}) = 4\pi \delta(\mathbf{r} - \mathbf{r}')$

Therefore, $\nabla \cdot \mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int 4\pi \delta(\mathbf{r} - \mathbf{r}') \rho(\mathbf{r}') d\tau = \frac{\rho(\mathbf{r})}{\epsilon_0}$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

The divergence of electric field is equal to the charge density divided by ϵ_0



**Thank
You**

